CHAPTER 03

Bonding

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: cd2a2d8d9b891a176ebd4ab753e3ca67cab25853ab8c6be586355672d171bdcf

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch03:exercise:quantum | quantum:stage-1, quantum:stage-2, quantum:stage-3, quantum:stage-4, quantum:stage-5, quantum:stage-6

ch03:exercise:bond-count | bond-count:stage-1, bond-count:stage-2, bond-count:stage-3

ch03:exercise:amide-resonance | amide-resonance:stage-1, amide-resonance:stage-2, amide-resonance:stage-3

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ch03:molecule:ethene | ethene

ch03:molecule:ethyne | ethyne

ch03:molecule:acetamide | acetamide

ch03:molecule:acetaldehyde | acetaldehyde

ch03:molecule:acetate | acetate

ch03:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 3.1 Atomic Orbitals and Quantum Numbers

scope-2: 3.2 Molecular Orbitals

scope-3: 3.3 Hybridization

Objective: Apply allowed quantum numbers

Objective: Count sigma and pi bonds

Objective: Explain hybridization and resonance

Supplied Kaplan curriculum: Chapter 3; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter03b.xhtml

OpenStax: quantum numbers, shapes and nodes | https://openstax.org/books/chemistry-atoms-first-2e/pages/3-3-development-of-quantum-theory

OpenStax: electronic structure, subshell energies and filling | https://openstax.org/books/chemistry-atoms-first-2e/pages/3-4-electronic-structure-of-atoms-electron-configurations

OpenStax: valence-bond overlap and relative bond strengths | https://openstax.org/books/chemistry/pages/8-1-valence-bond-theory

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

orbitals | Atomic orbitals and quantum numbers

Orbitals describe probability, not paths

orbitals:rule | Core Rules

n = 1,2,… sets the shell. l = 0,…,n−1 sets the subshell (s,p,d,f). m_l = −l,…,+l labels orbitals; m_s = ±1/2 labels electron spin. Each orbital holds at most two electrons with opposite spins. In the usual multielectron-atom model, subshell energies within one shell increase s < p < d < f (for subshells allowed in that shell). In the nonrelativistic hydrogen-like model, orbitals with the same n have equal energy; do not apply the multielectron splitting to that model.

Visual card: Quantum Numbers (n, l, m_l, m_s), Nodal Planes & Ground-State Electron Configurations

Chemistry diagram: Isolated carbon-atom spin occupancy model, not a molecular structure or an energy-scale diagram.

Total Nodes = n − 1 | Angular Nodes = l | Radial Nodes = n − l − 1

Total Nodes = n − 1

Angular Nodes = l

Radial Nodes = n − l − 1

Cr ($4s^1 3d^5$) & Cu ($4s^1 3d^{10}$)

s Orbital (l = 0) | Spherical (0 angular nodes) | 1 orbital → max 2 e^-

p Orbital (l = 1) | Dumbbell (1 nodal plane) | 3 orbitals → max 6 e^-

d Orbital (l = 2) | Cloverleaf (2 nodal planes) | 5 orbitals → max 10 e^-

Aufbau, Pauli & Hund Rules — — Fill lowest energy first (n+l rule); max 2 e^- per orbital with opposite spins; fill degenerate orbitals singly with parallel spins.

Cation Formation & d-Block Exceptions — — Remove electrons from HIGHEST n shell first (4s before 3d in Fe^{2+}); remember Cr ([Ar] $4s^1 3d^5$) and Cu ([Ar] $4s^1 3d^{10}$).

orbitals:cue | Memory Cues

One s orbital; three p orbitals; five d orbitals; seven f orbitals. A p subshell can hold six electrons. For hydrogen-like atomic orbitals: total nodes = n−1; angular nodes = l; radial nodes = n−l−1. A 3p orbital has one angular node and one radial node.

Visual card: Quantum Number & Ion Electron Configuration Checklist

Which atomic orbital rule are you applying?

Counting Nodes in Any Orbital (e.g., 3p) — Total = n−1 (2); Angular = l (1); Radial = n−l−1 (1) — l = 0 (s), 1 (p), 2 (d), 3 (f). Angular nodes are flat planes/cones; radial nodes are spherical shells.

Forming a Transition-Metal Cation (e.g., Fe^{2+}) — Empty 4s BEFORE 3d! — Neutral Fe is [Ar] $4s^2 3d^6$, so Fe^{2+} loses its two 4s electrons first → [Ar] $3d^6$ (4 unpaired electrons, paramagnetic).

Checking Pauli vs. Hund Violation — Same 4 Quantum Numbers = Pauli; Paired Too Early = Hund — Two arrows ↑↑ in the same box violates Pauli Exclusion; pairing ↑↓ in px while py is empty is an excited state (Hund).

orbitals:trap | Common Mistakes / MCAT Traps

Orbital phase signs are wavefunction signs, not positive and negative electric charges. Electrons do not orbit like planets. s orbitals have spherical angular symmetry; a p orbital has two lobes separated by a nodal plane. Higher-n orbitals may also have radial nodes. A node is a zero of the wavefunction, not a place where an electron waits.

Visual card: Removing 3d Electrons Before 4s in Transition Metal Ions

Common Trap — — Writing Fe^{2+} as [Ar] $4s^2 3d^4$ by subtracting from 3d instead of the outermost n=4 shell.

Correct Reasoning — — Once 3d orbitals are occupied, 4s is higher in energy and furthest from the nucleus: always ionize 4s electrons first!

orbitals:app | MCAT Applications

Carbon bonding uses valence orbitals from the second shell. Hybridization is a model for bonding and geometry, not a change in the allowed quantum numbers.

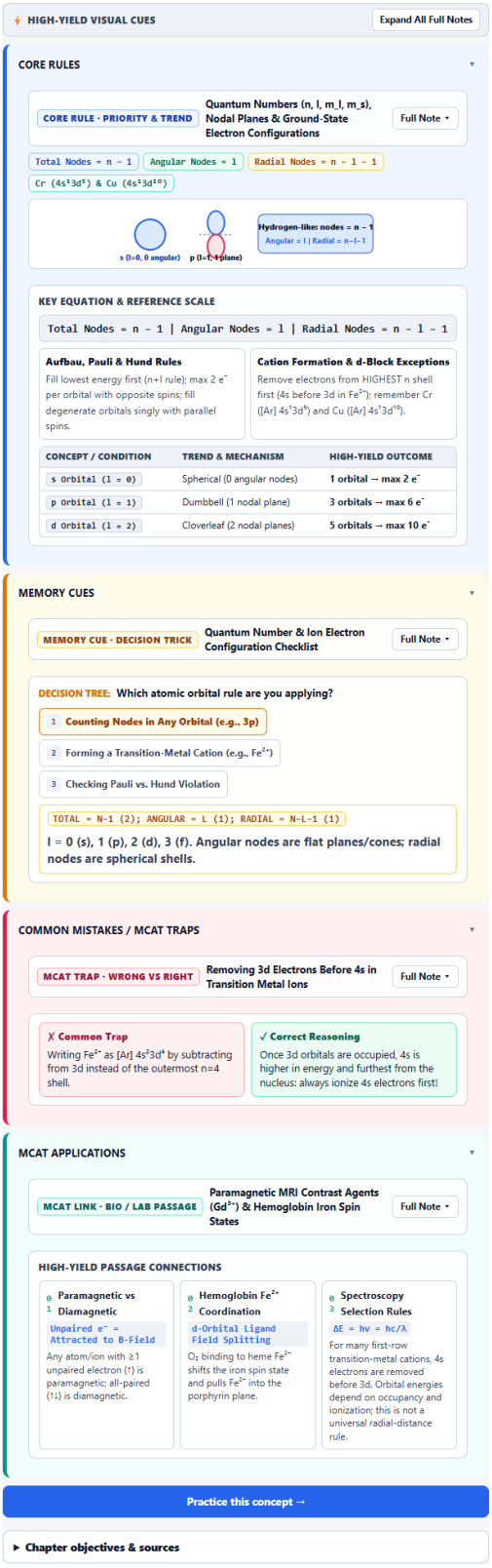
Visual card: Paramagnetic MRI Contrast Agents (Gd^{3+}) & Hemoglobin Iron Spin States

Paramagnetic vs Diamagnetic — Unpaired e^- = Attracted to B-Field — Any atom/ion with ≥1 unpaired electron (↑) is paramagnetic; all-paired (↑↓) is diamagnetic.

Hemoglobin Fe^{2+} Coordination — d-Orbital Ligand Field Splitting — O_2 binding to heme Fe^{2+} shifts the iron spin state and pulls Fe^{2+} into the porphyrin plane.

Spectroscopy Selection Rules — $\Delta E = h\nu = hc/\lambda$ — For many first-row transition-metal cations, 4s electrons are removed before 3d. Orbital energies depend on occupancy and ionization; this is not a universal radial-distance rule.

Screenshot: orbitals-visuals



orbitals:step-1 | 1. Count valence electrons

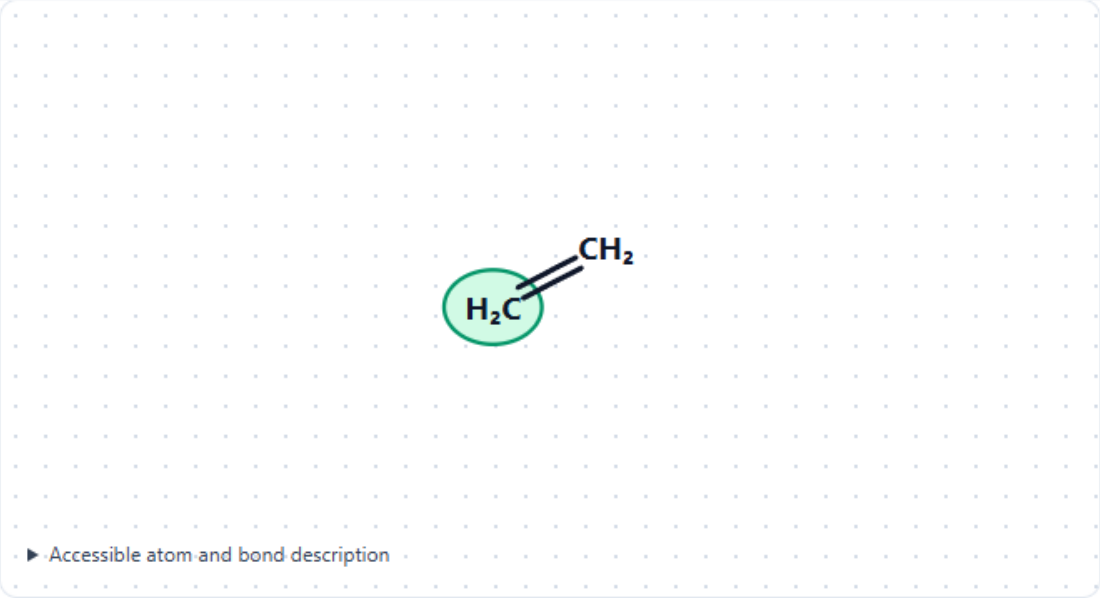
The molecular drawing is ethene, $H_2C{=}CH_2$, shown as a later bonding example. The separate box diagram describes an isolated neutral carbon atom: $1s^2 2s^2 2p^2$, with four valence electrons in the n=2 shell. Do not interpret the C=C lines as

orbital boxes or apply the isolated-atom occupancy directly to a bonded carbon.

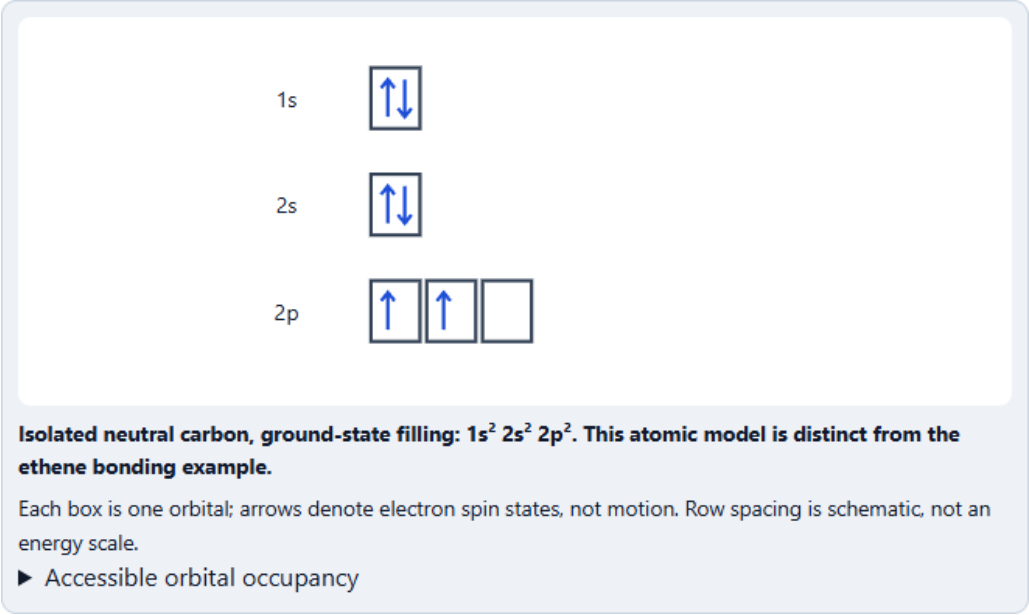
Displayed molecule ID: ethene

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Screenshot: orbitals-step-1



Screenshot: orbitals-step-1-diagram



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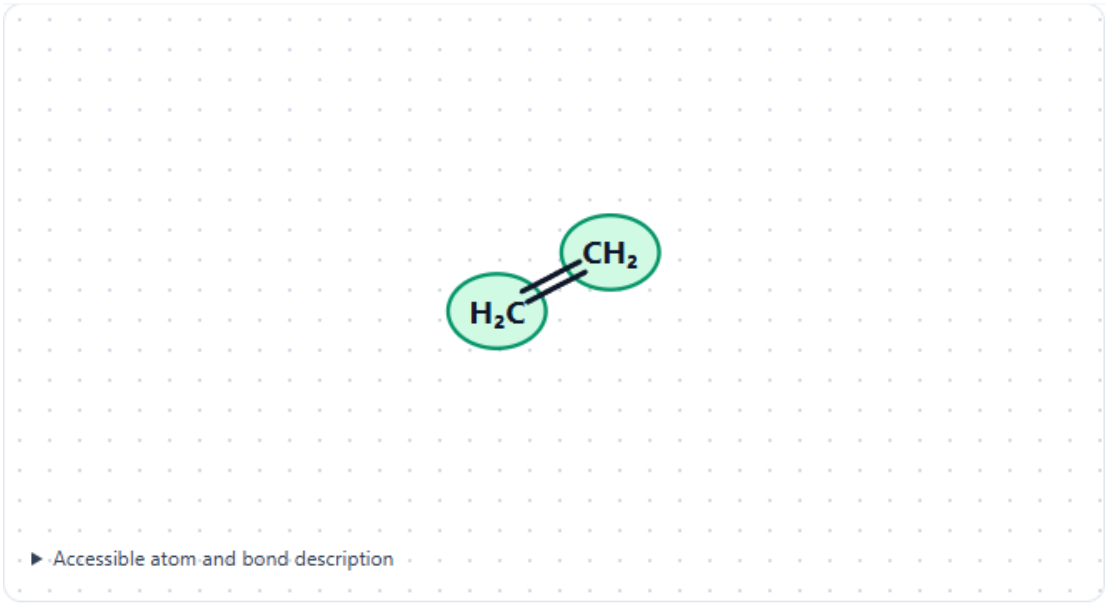
orbitals:step-2 | 2. Fill degenerate orbitals

In the isolated-carbon box diagram, the two 2p electrons occupy separate degenerate orbitals with parallel spins (Hund's rule). Electrons paired in a single orbital have opposite spins (Pauli exclusion). Ethene is the molecular example above; its bonding orbitals are a different model, developed in the next lesson.

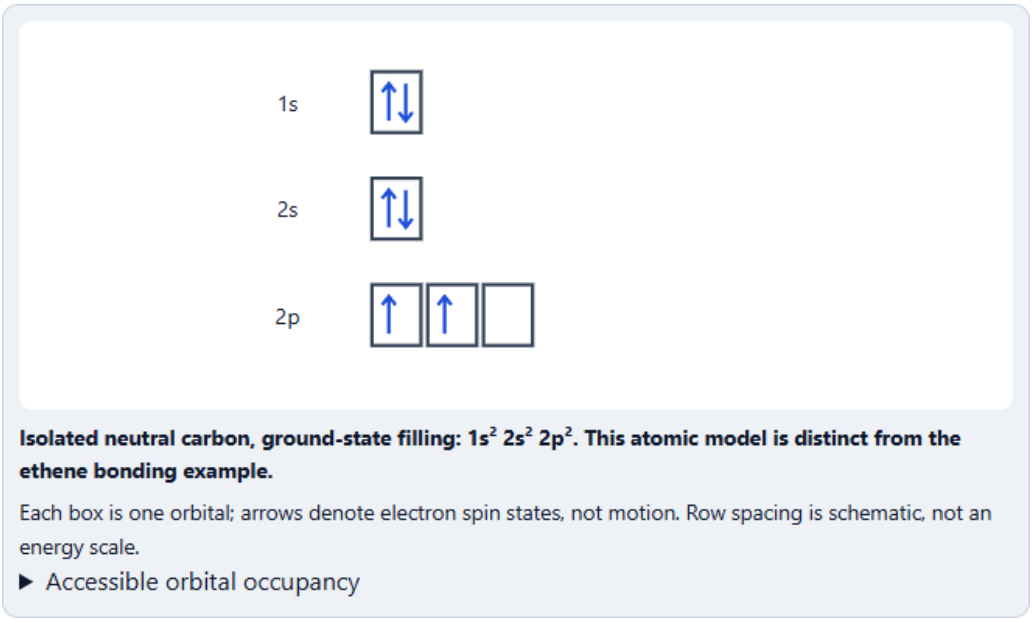
Displayed molecule ID: ethene

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Screenshot: orbitals-step-2



Screenshot: orbitals-step-2-diagram



{"diagramSpecification": {"type": "orbitals", "caption": "Isolated neutral carbon, ground-state filling: $1s^2\ 2s^2\ 2p^2$. This atomic model is distinct from the ethene bonding example.", "subshells": [{"label": "1s", "electrons": [2]}, {"label": "2s", "electrons": [2]}, {"label": "2p", "electrons": [1, 1, 0]}]}}

sigma-pi | Sigma, pi, and molecular orbitals

End-on vs. side-on overlap

sigma-pi:rule | Core Rules

A sigma bond has electron density along the internuclear axis. A pi bond comes from side-on p-orbital overlap. A double bond has one sigma and one pi; a triple bond has one sigma and two pi. Constructive overlap forms a lower-energy bonding MO; destructive overlap gives an antibonding MO with an added node. For comparable carbon–carbon bonds, an individual pi contribution is generally weaker than the sigma contribution because side-on overlap is less effective. Nevertheless, a double bond and a triple bond are generally stronger and shorter overall than a single bond. Their strength increments are unequal; do not add universal, context-free sigma and pi energies.

Visual card: Sigma (σ) vs. Pi (π) Overlap, Bond Order & Molecular Orbital (MO) Theory

Chemistry diagram: Bonding-orbital overlap schematic. Lobe colors denote phase, not electric charge or electron paths.

Bond Order = ½ (Bonding e^- − Antibonding e^-*) | Double Bond = 1σ + 1π | Triple Bond = 1σ + 2π

Single = 1σ

Double = 1σ + 1π

Triple = 1σ + 2π

Bond Order = ½(e_b − e_a*)

σ Bond (Head-on) | Cylindrical symmetry along axis | Stronger overlap; free rotation

π Bond (Side-by-side) | Electron density above/below nodal plane | Locks rotation; nucleophilic site

π* Antibonding MO | Destructive out-of-phase overlap | Adds node between nuclei; weakens bond

Head-On (σ) vs. Side-by-Side (π) — — Head-on σ overlap concentrates electron density directly between nuclei (stronger individual bond) and permits 360° rotation.

Antibonding Orbitals (σ*, π*) — — Populating an antibonding orbital creates a nodal plane between the nuclei, lowering bond order and lengthening the bond.

sigma-pi:cue | Memory Cues

Bond order = (bonding electrons − antibonding electrons)/2 in the simple MO model. More bond order usually means shorter, stronger bonds for a given atom pair.

Visual card: Fast Sigma & Pi Bond Counting Formula for Any Graph

What bond property are you comparing?

Counting Total σ and π Bonds — Every Atom-Atom Link = 1σ; Extra Lines = π — Remember to count implicit C–H σ bonds! A double bond adds +1π; a triple bond adds +2π.

Individual σ vs. π Bond Strength — 1 σ Bond > 1 π Bond (Head-On > Parallel) — The π bond is weaker and higher in energy (HOMO), making alkenes nucleophilic in addition reactions.

Total Bond Energy & Bond Length — C≡C > C=C > C–C Strength; C≡C < C=C < C–C Length — Higher bond order (3 > 2 > 1) pulls nuclei closer together and requires more total energy to cleave completely.

sigma-pi:trap | Common Mistakes / MCAT Traps

A pi bond restricts rotation because twisting disrupts overlap. Bond rotation is possible only by paying an energy cost; do not imagine a double bond freely rotating at room temperature.

Visual card: Forgetting Implicit C–H Bonds When Counting Total σ Bonds

Common Trap — — Counting only the visible C–C skeleton lines in skeletal structures and missing all the C–H σ bonds.

Correct Reasoning — — Expand every carbon to 4 total bonds first (or use Total σ = Total Atoms − 1 + Rings) so every C–H σ bond is included.

sigma-pi:app | MCAT Applications

Carbonyl addition converts a pi bond into sigma bonds. Following sigma and pi electrons helps explain the change from planar to tetrahedral geometry.

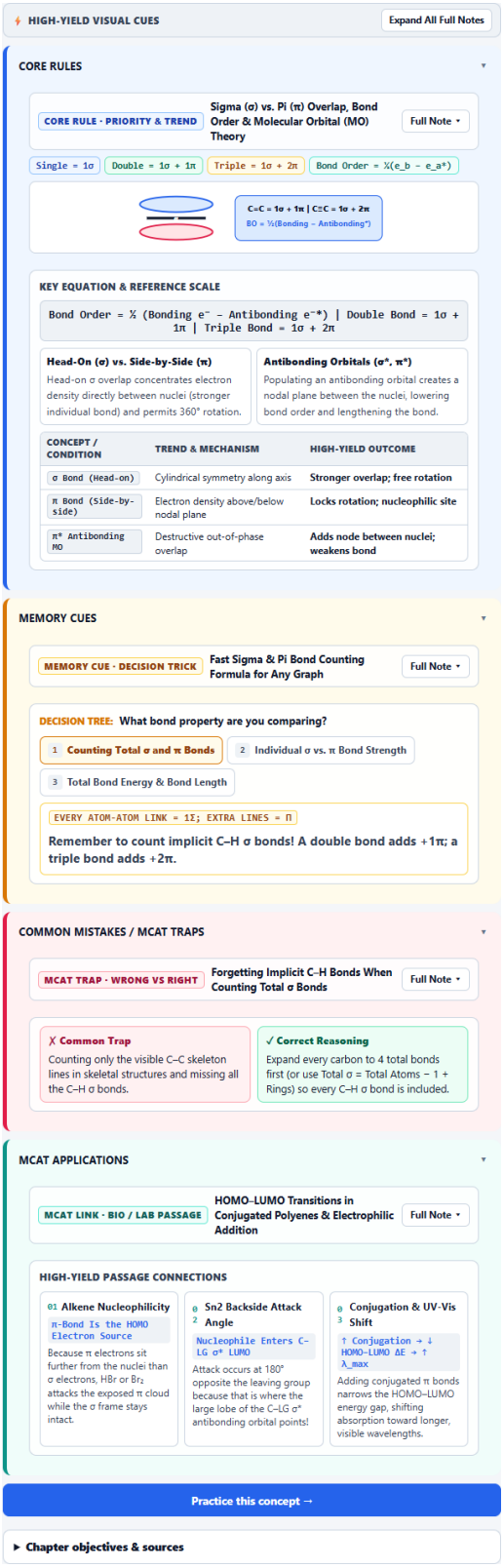
Visual card: HOMO–LUMO Transitions in Conjugated Polyenes & Electrophilic Addition

Alkene Nucleophilicity — π-Bond Is the HOMO Electron Source — Because π electrons sit further from the nuclei than σ electrons, HBr or Br_2 attacks the exposed π cloud while the σ frame stays intact.

Sn2 Backside Attack Angle — Nucleophile Enters C–LG σ* LUMO — Attack occurs at 180° opposite the leaving group because that is where the large lobe of the C–LG σ* antibonding orbital points!

Conjugation & UV-Vis Shift — ↑ Conjugation → ↓ HOMO–LUMO ΔE → ↑ λ_max — Adding conjugated π bonds narrows the HOMO–LUMO energy gap, shifting absorption toward longer, visible wavelengths.

Screenshot: sigma-pi-visuals



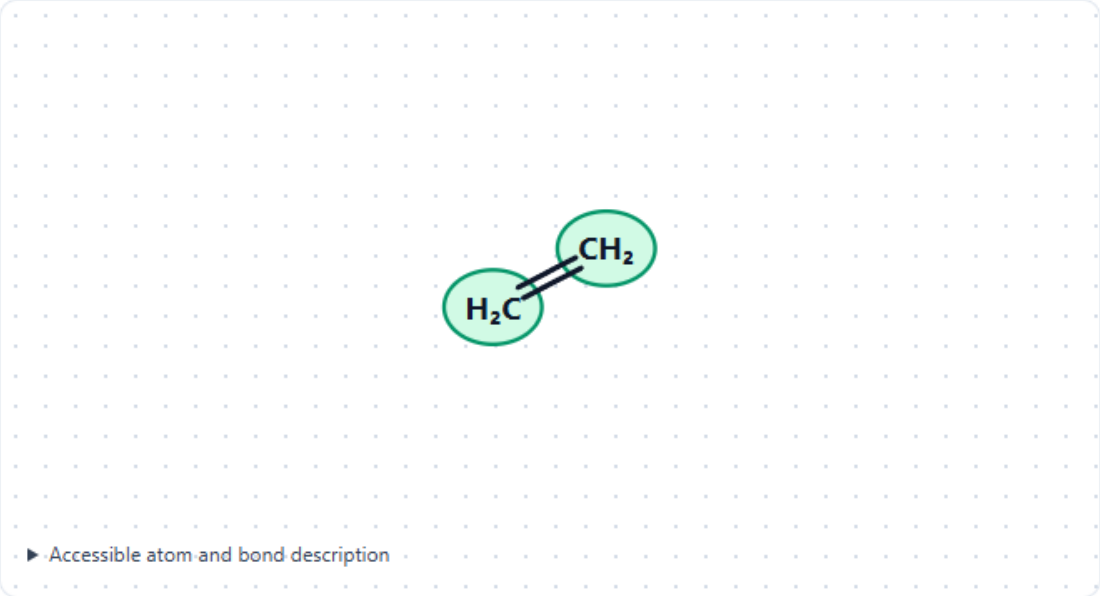
sigma-pi:step-1 | 1. Double bond inventory

Ethene has one C–C sigma bond, one C–C pi bond, and four C–H sigma bonds: five sigma and one pi overall. The H_2C and CH_2 labels show the hydrogens; the four C–H bonds are not drawn as separate lines.

Displayed molecule ID: ethene

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Screenshot: sigma-pi-step-1



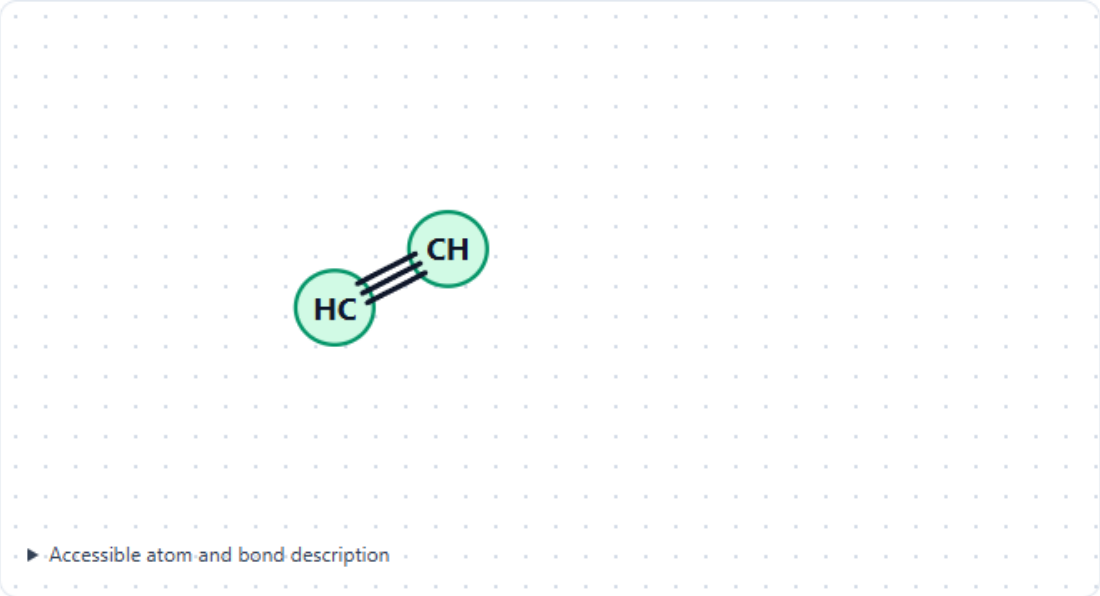
sigma-pi:step-2 | 2. Triple bond inventory

Ethyne has one C–C sigma plus two perpendicular pi bonds. Each CH label includes one hydrogen whose C–H bond is not drawn separately. Add those two C–H sigma bonds: three sigma and two pi overall.

Displayed molecule ID: ethyne

{"highlights": {"prin": [0, 1]}, "revealName": true}

Screenshot: sigma-pi-step-2



hybridization | Hybridization, geometry, and resonance

Count domains, then check conjugation

hybridization:rule | Core Rules

Typical tetrahedral carbon is sp^3 (about 109.5°), trigonal planar carbon is sp^2 (about 120°), and linear carbon is sp (180°). The s character is 25%, 33%, and 50%, respectively. Resonance contributors differ only in electron placement, not atom connectivity.

Visual card: Hybridization, % s-Character, Bond Lengths & Resonance Delocalization Rules

Chemistry diagram: Conjugated amide and carboxylate examples. A flat drawing does not measure bond angles.

sp^3 (25% s, 109.5°)

sp^2 (33% s, 120°)

sp (50% s, 180°)

Amide/Aniline N = sp^2 Planar!

sp^3 (4 domains) | 25% s-character · 109.5° | Longest C–H; pKa ~50 (least acidic)

sp^2 (3 domains) | 33% s-character · 120° | Planar + 1 unhybridized p; pKa ~44

sp (2 domains) | 50% s-character · 180° | Shortest/strongest C–H; pKa ~25 (acidic!)

Count Steric Number (σ Bonds + Localized Lone Pairs) — — 4 domains = sp^3 (109.5° tetrahedral); 3 domains = sp^2 (120° trigonal planar); 2 domains = sp (180° linear).

Resonance Hybridization Override — — A lone pair adjacent to a π bond (amides, esters, aniline, pyrrole) rehybridizes from sp^3 → sp^2 to sit in an unhybridized p orbital!

Percent s-Character Trend — — sp (50% s) > sp^2 (33% s) > sp^3 (25% s): higher s-character holds electrons closer to the nucleus → shorter bonds & stronger acidity.

hybridization:cue | Memory Cues

For ordinary carbon: four electron domains → sp^3; three → sp^2; two → sp. A multiple bond counts as one domain.

Visual card: Resonance & Lone-Pair Hybridization Detector

Does the heteroatom lone pair participate in resonance?

Next to a π Bond (Amide, Aniline, Ester) — Rehybridizes to sp^2 (Lone Pair in p Orbital!) — Delocalizing the lone pair into the adjacent π system lowers energy, making amide C–N bonds planar and non-basic.

Atom ALREADY Has a π Bond (Pyridine N) — Lone Pair Stays in sp^2 Hybrid Orbital! — Because pyridine nitrogen already uses its p orbital for the aromatic ring π bond, its lone pair sits orthogonal in an sp^2 orbital (basic!).

No Adjacent π System (Alkyl Amine RNH_2) — Standard sp^3 (Trigonal Pyramidal ~107°) — Localized lone pair occupies an sp^3 hybrid orbital and is freely available to accept a proton.

hybridization:trap | Common Mistakes / MCAT Traps

Do not mechanically label an amide nitrogen sp^3 just because it has three sigma bonds and a lone pair. Its lone pair occupies a p orbital conjugated with the carbonyl, making the amide approximately planar.

Visual card: Calling Amide Nitrogen sp^3 or Moving Atoms Between Resonance Structures

Common Trap — — Counting 3 σ bonds + 1 lone pair on an amide nitrogen and calling it sp^3 trigonal pyramidal.

Correct Reasoning — — Amide, ester, and aniline heteroatoms are sp^2 trigonal planar because their lone pair delocalizes into the adjacent π system!

hybridization:app | MCAT Applications

Peptide-bond resonance restricts C–N rotation. Delocalized charge explains carboxylate stability and why its two C–O bonds are equivalent in the resonance hybrid.

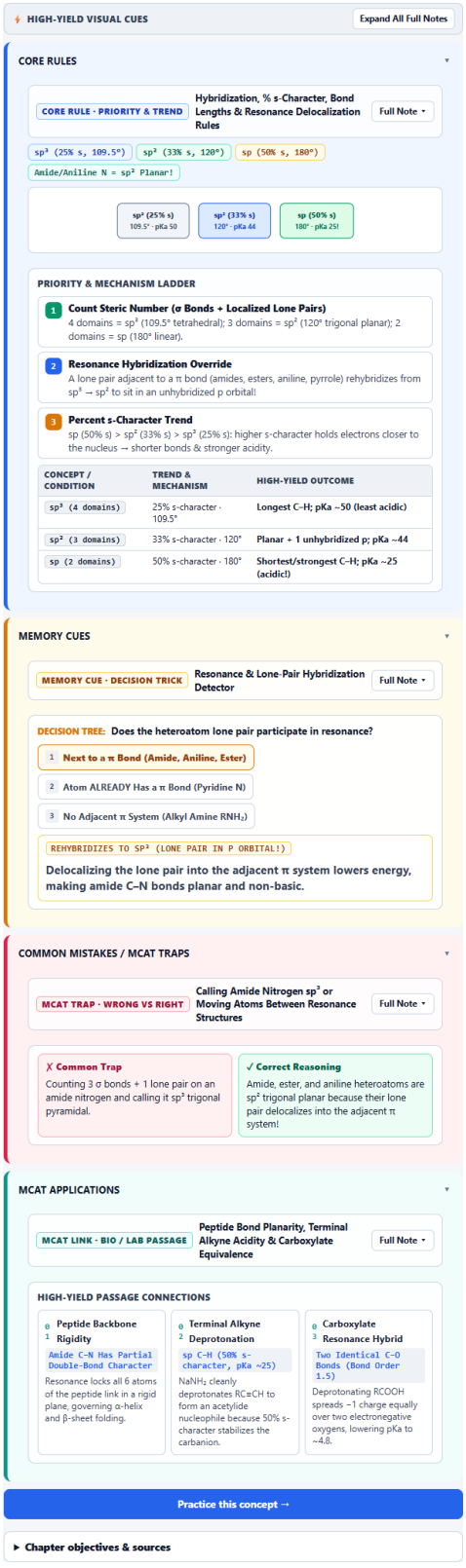
Visual card: Peptide Bond Planarity, Terminal Alkyne Acidity & Carboxylate Equivalence

Peptide Backbone Rigidity — Amide C–N Has Partial Double-Bond Character — Resonance locks all 6 atoms of the peptide link in a rigid plane, governing α-helix and β-sheet folding.

Terminal Alkyne Deprotonation — sp C–H (50% s-character, pKa ~25) — $NaNH_2$ cleanly deprotonates RC≡CH to form an acetylide nucleophile because 50% s-character stabilizes the carbanion.

Carboxylate Resonance Hybrid — Two Identical C–O Bonds (Bond Order 1.5) — Deprotonating RCOOH spreads −1 charge equally over two electronegative oxygens, lowering pKa to ~4.8.

Screenshot: hybridization-visuals



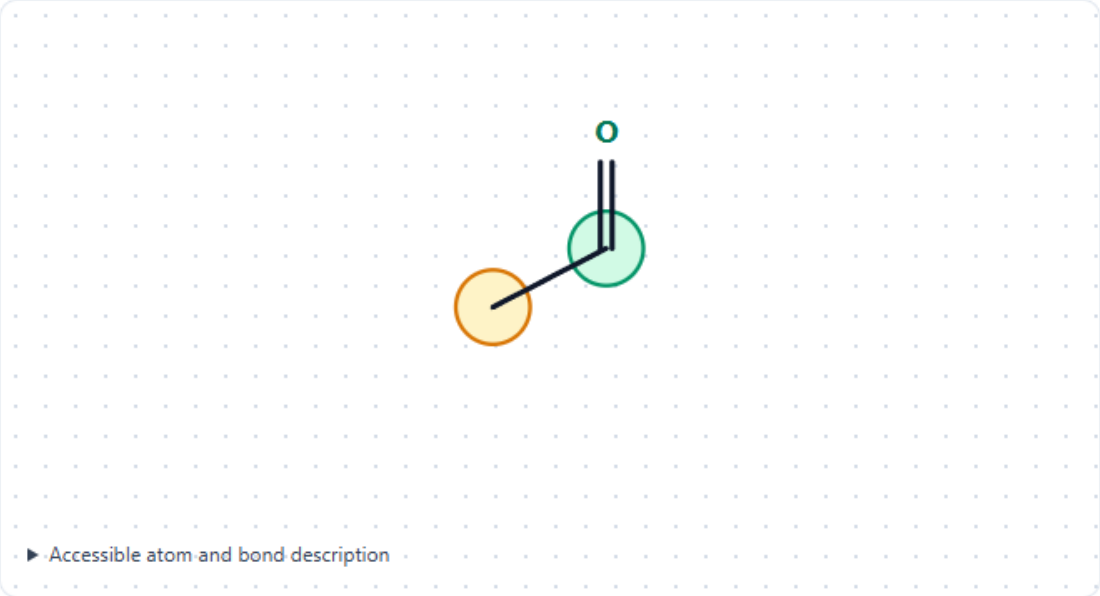
hybridization:step-1 | 1. Carbonyl geometry

The carbonyl carbon has three sigma-bond directions and a p orbital for the pi bond: sp^2 and approximately planar. Its methyl neighbor is sp^3.

Displayed molecule ID: acetaldehyde

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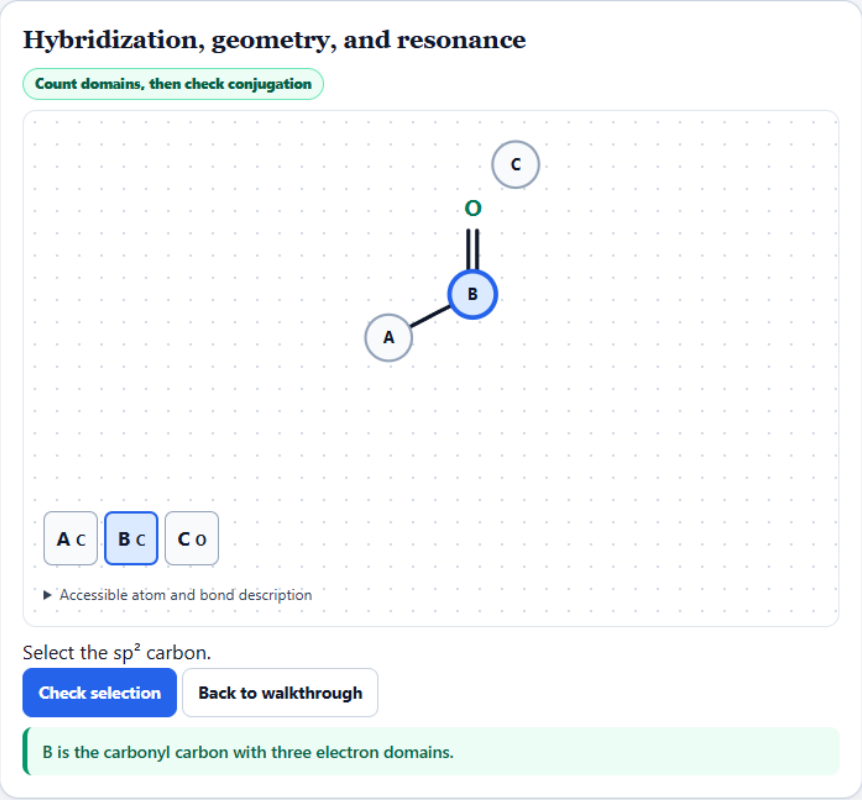
Screenshot: hybridization-step-1



Embedded challenge: Select the sp^2 carbon.

{"prompt": "Select the sp^2 carbon.", "answer": [1], "explanation": "B is the carbonyl carbon with three electron domains."}

Screenshot: hybridization-step-1-challenge



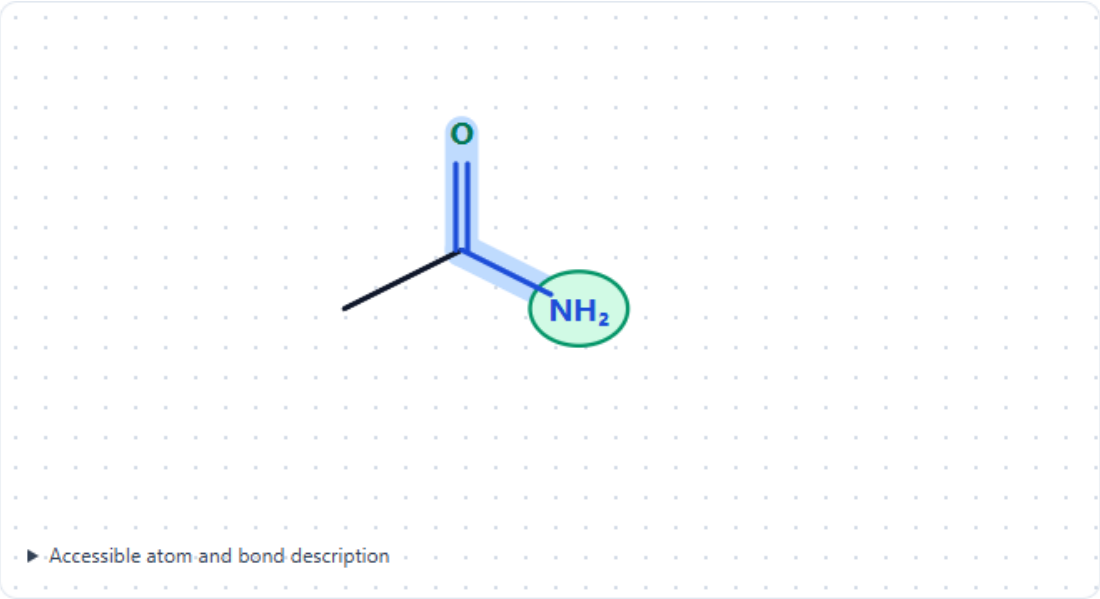
hybridization:step-2 | 2. Amide conjugation

Nitrogen donates lone-pair electron density into the carbonyl system. The C–N bond has partial double-bond character; amide nitrogen is much less basic than a typical amine.

Displayed molecule ID: acetamide

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Screenshot: hybridization-step-2



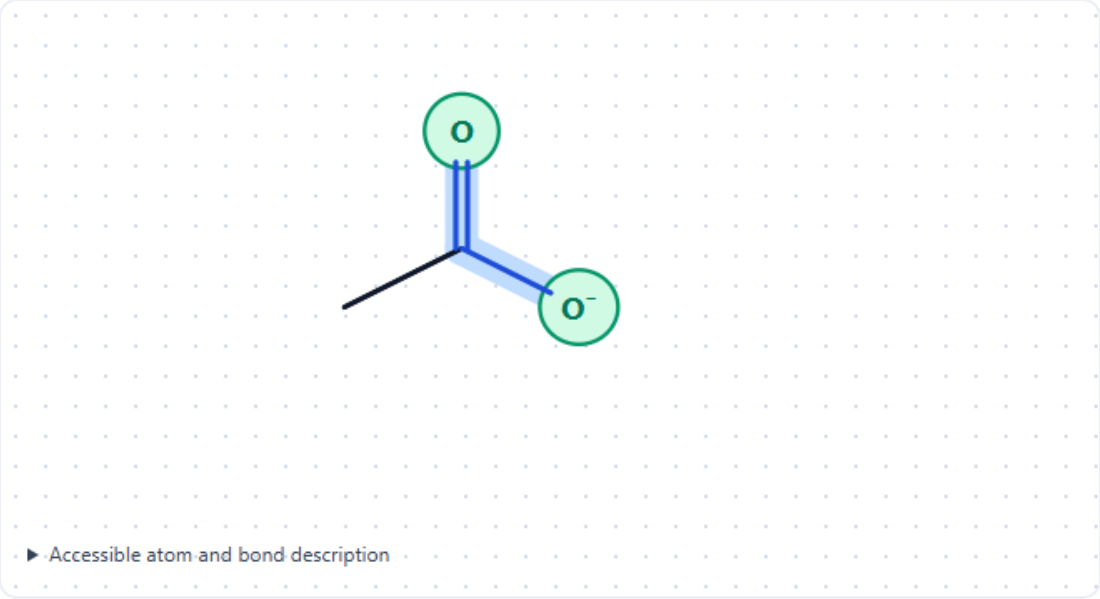
hybridization:step-3 | 3. Resonance hybrid

This drawing is one contributor of acetate. The other exchanges the two oxygen bond orders and the negative charge. The actual ion has delocalized charge; it does not flip back and forth between drawings.

Displayed molecule ID: acetate

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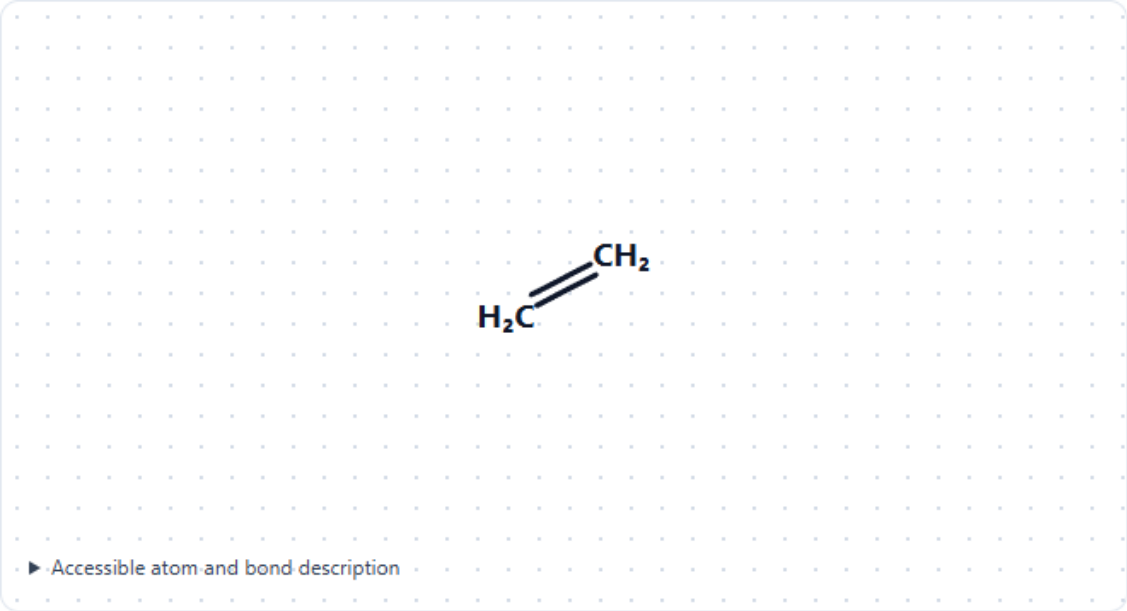
Screenshot: hybridization-step-3



quantum | Check allowed quantum numbers

The ethene drawing is a companion bonding example only. These questions concern atomic orbitals and subshells, not the total electrons or molecular orbitals of C2H4. No atomic filling diagram is supplied during this workshop.

Screenshot: quantum-stage-1-molecule



quantum:stage-1 | For a 2p electron, which n and l pair is allowed?

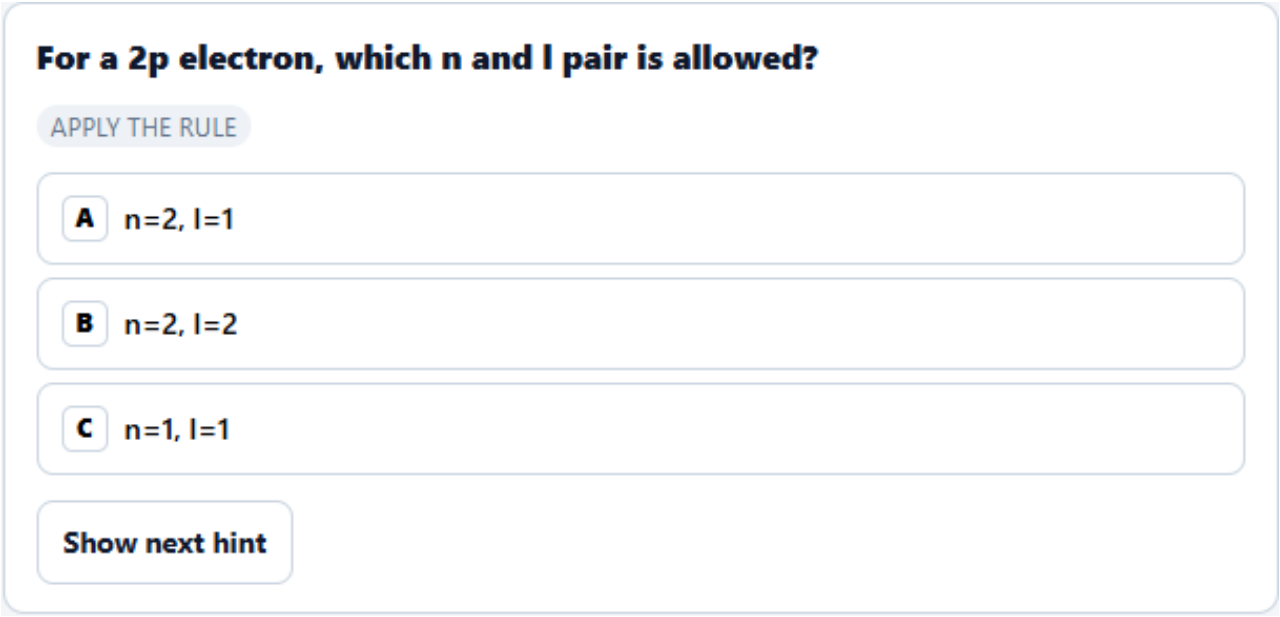
Answer type: choice

Option 0 (A): n=2, l=1

Option 1 (B): n=2, l=2

Option 2 (C): n=1, l=1

Screenshot: quantum-stage-1-question



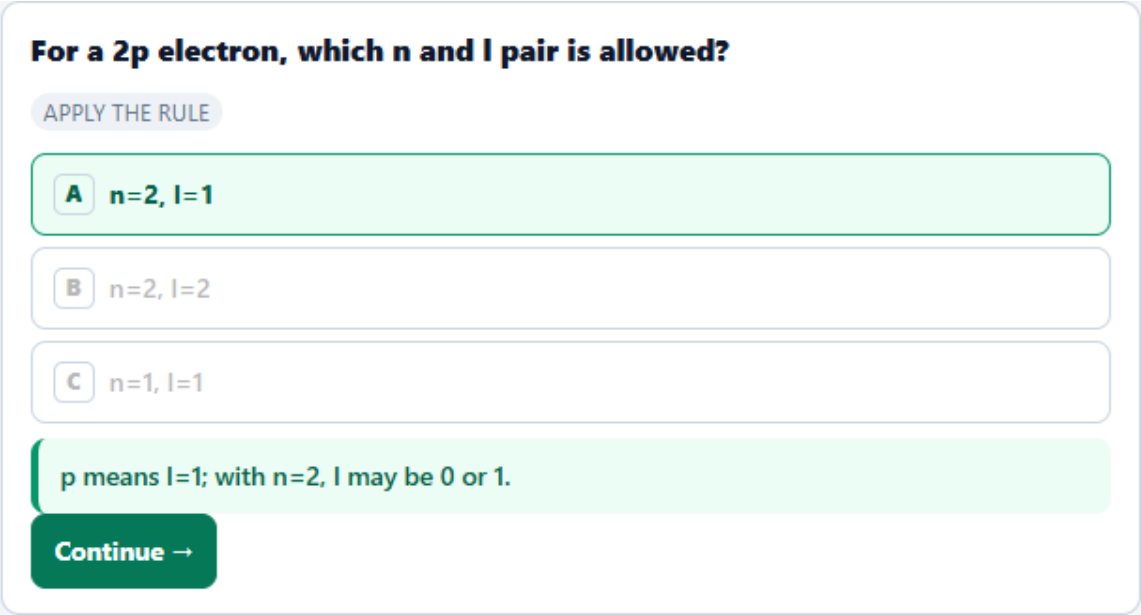
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use l ≤ n−1.

Hint 2: p is the subshell after s.

Explanation: p means l=1; with n=2, l may be 0 or 1.

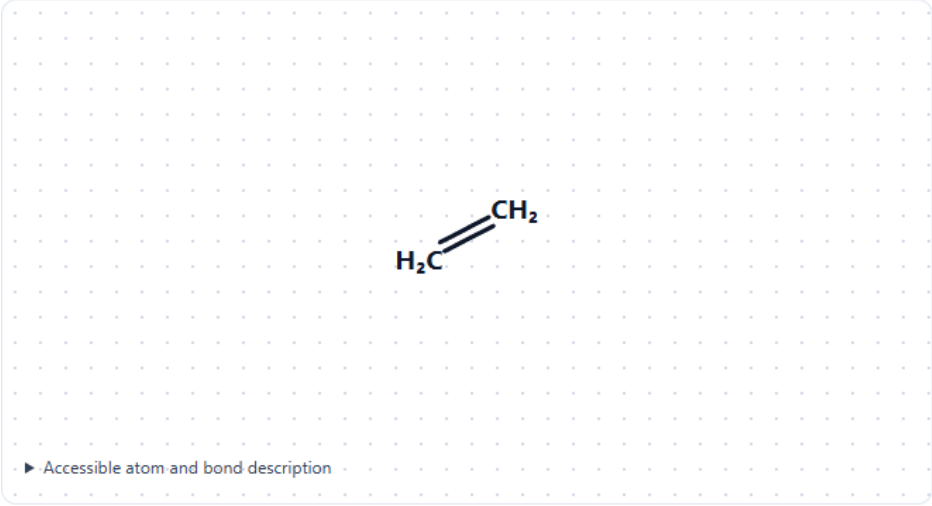
Screenshot: quantum-stage-1-feedback



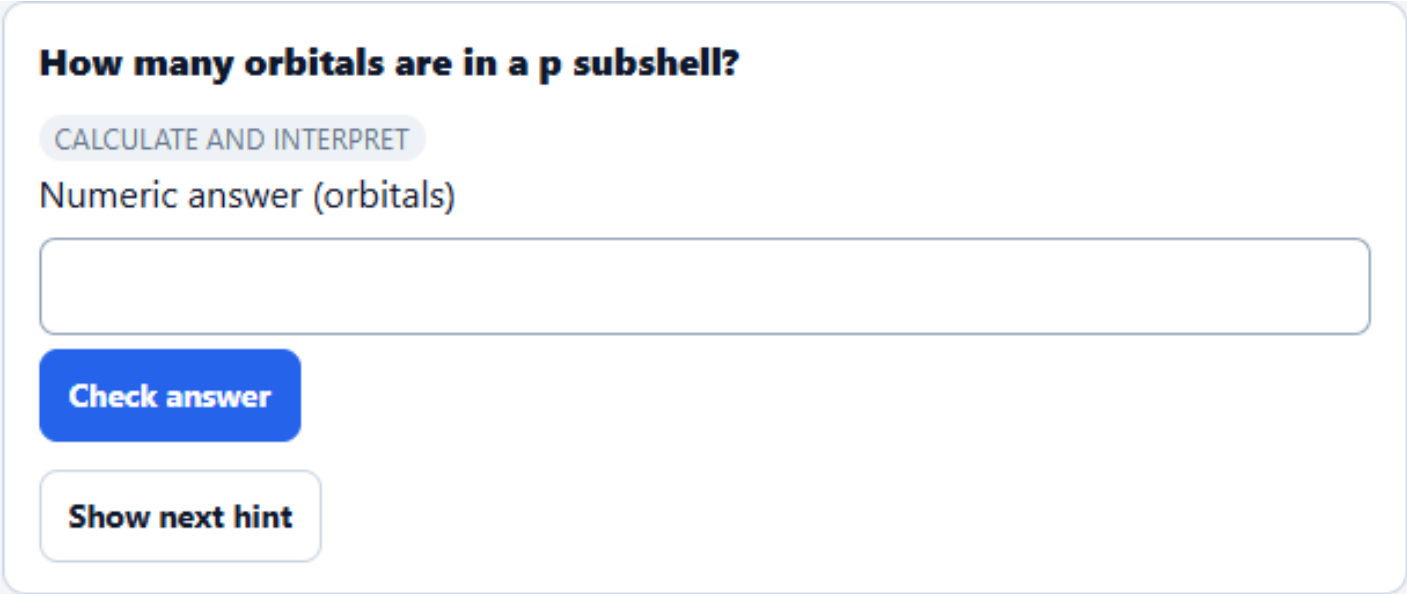
quantum:stage-2 | How many orbitals are in a p subshell?

Answer type: number | unit: orbitals

Screenshot: quantum-stage-2-molecule



Screenshot: quantum-stage-2-question



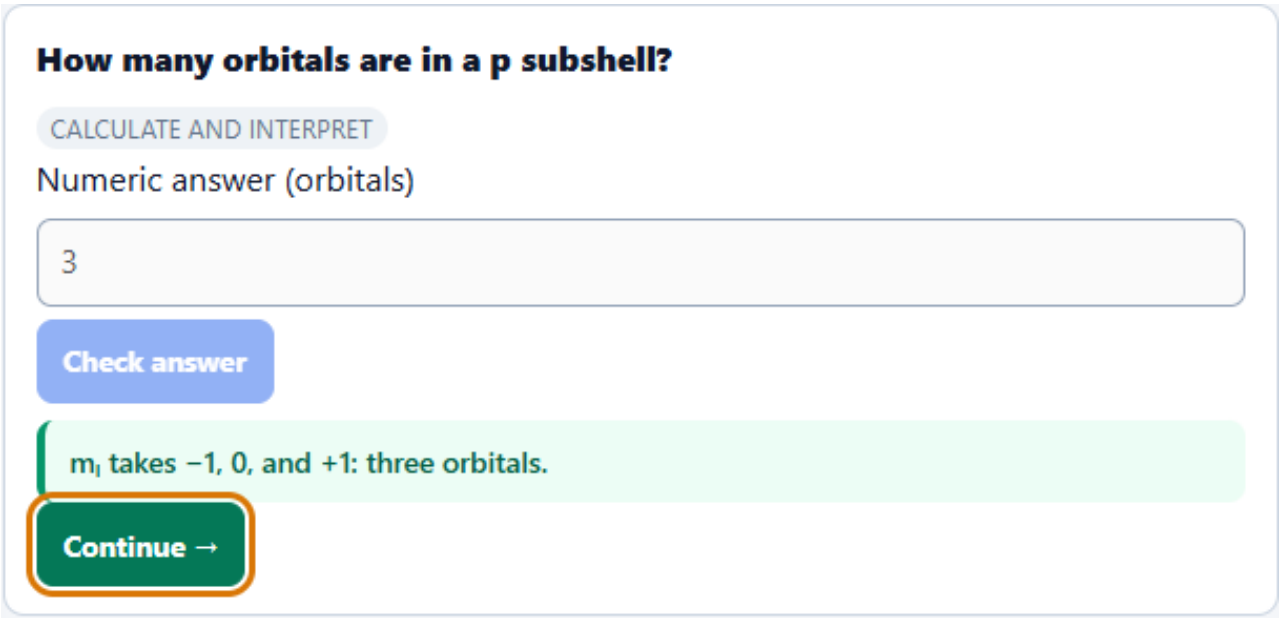
{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count the integers from −l to +l.

Hint 2: For p, l=1.

Explanation: m_l takes −1, 0, and +1: three orbitals.

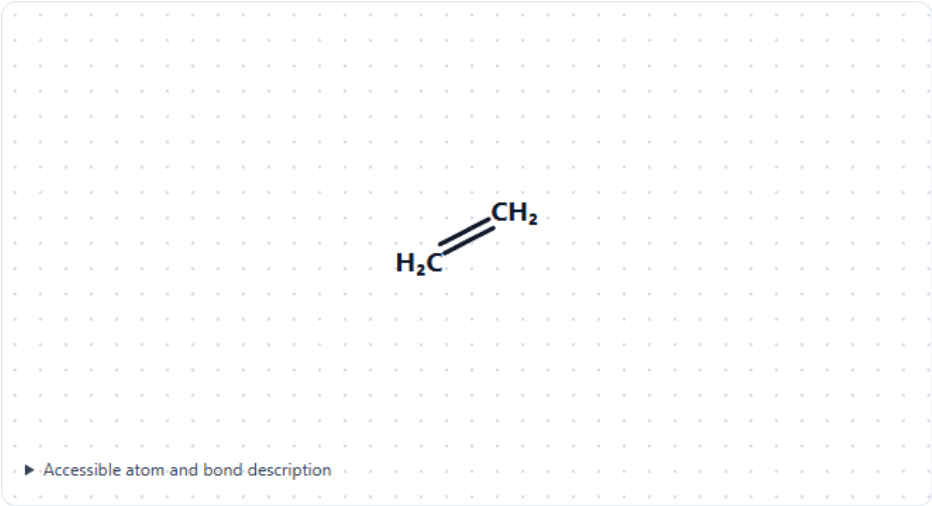
Screenshot: quantum-stage-2-feedback



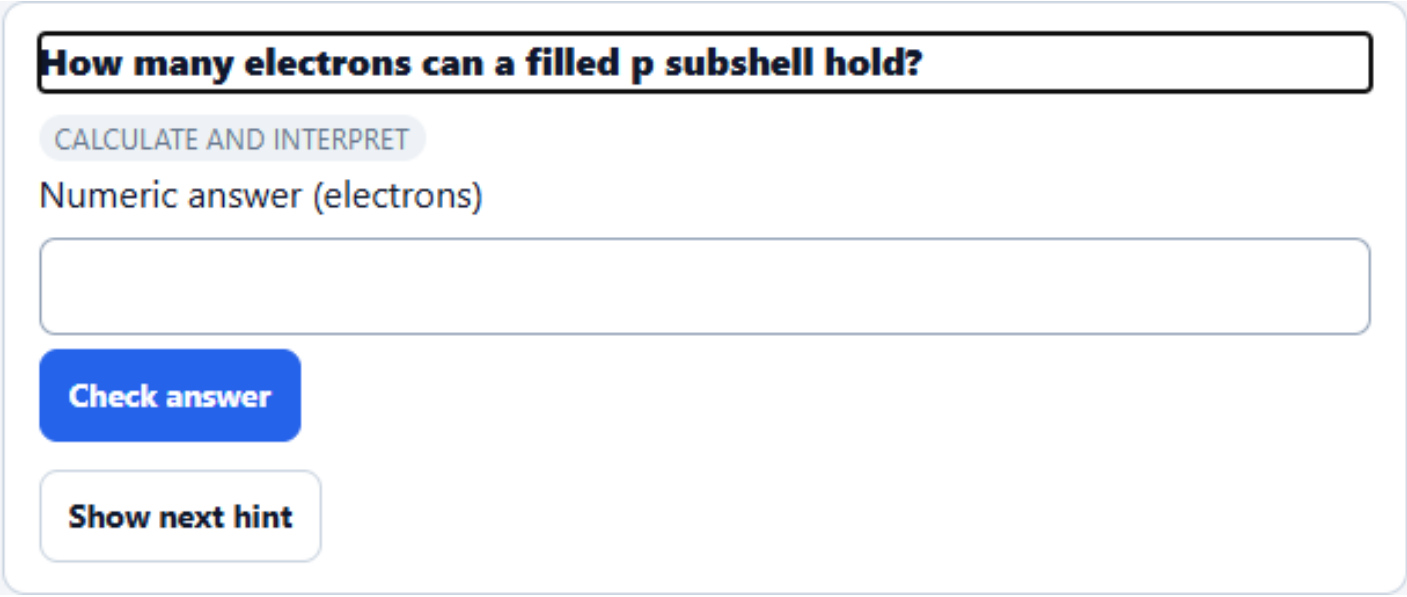
quantum:stage-3 | How many electrons can a filled p subshell hold?

Answer type: number | unit: electrons

Screenshot: quantum-stage-3-molecule



Screenshot: quantum-stage-3-question



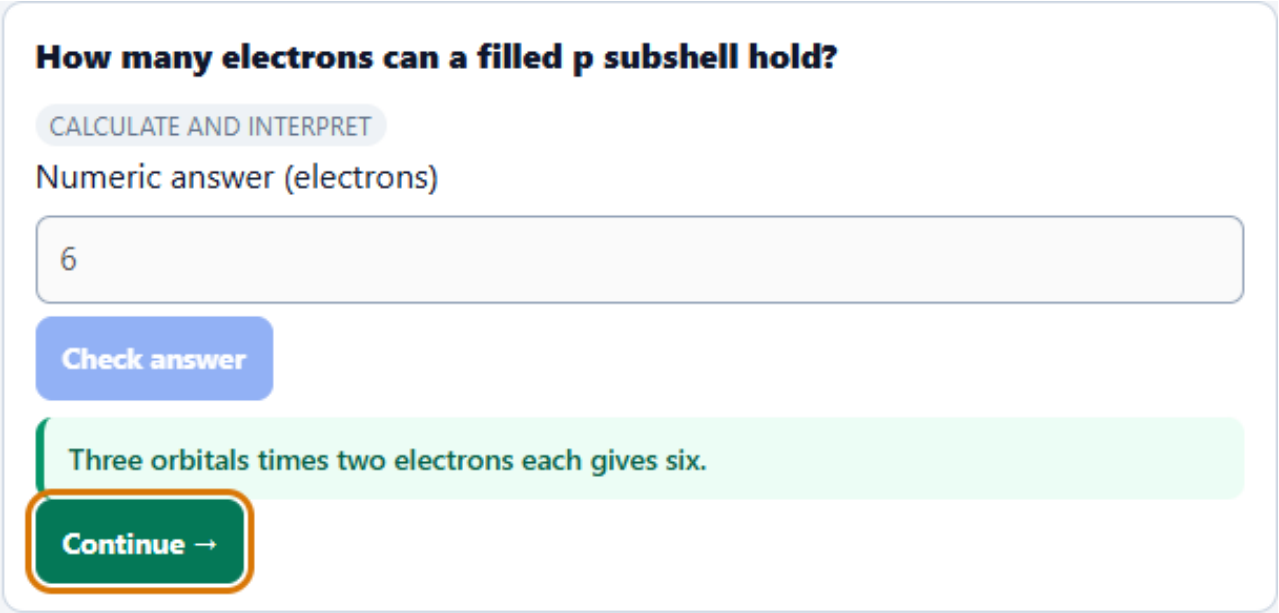
{"authorAnswer": 6, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use Pauli exclusion.

Hint 2: Each orbital holds two opposite-spin electrons.

Explanation: Three orbitals times two electrons each gives six.

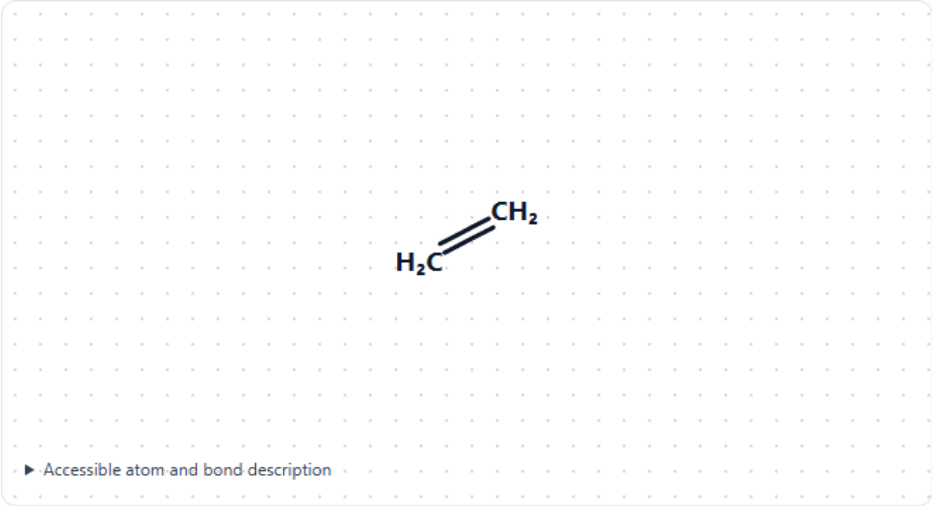
Screenshot: quantum-stage-3-feedback



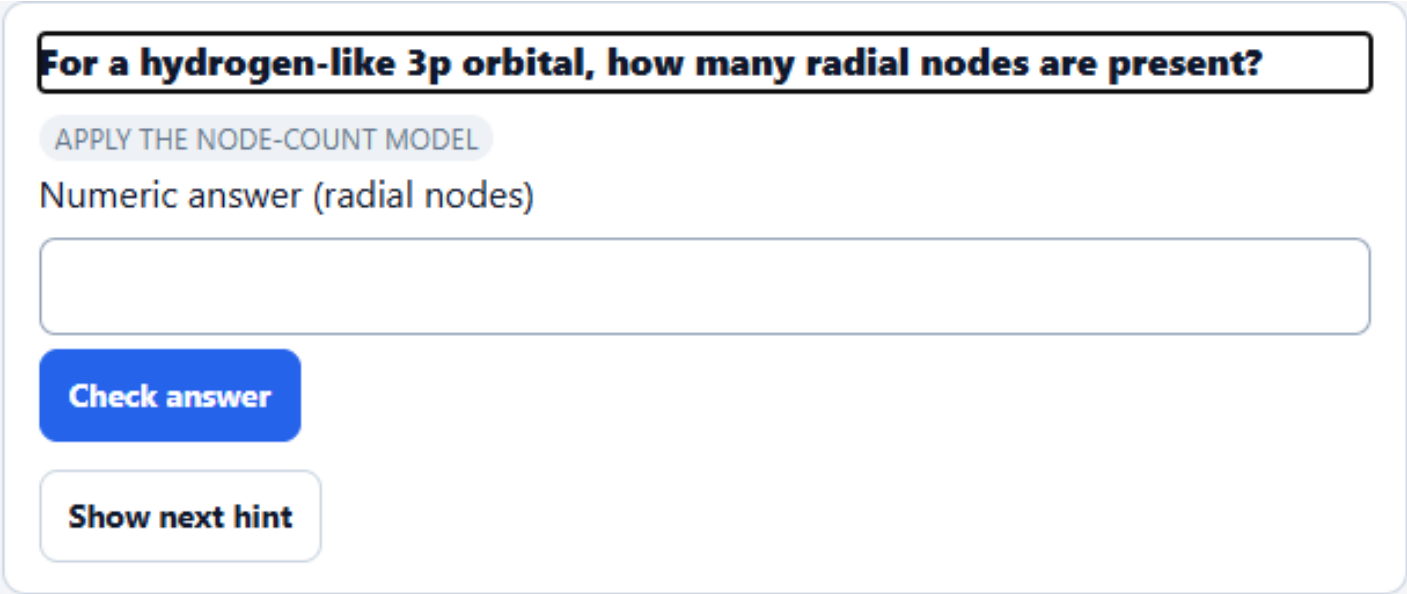
quantum:stage-4 | For a hydrogen-like 3p orbital, how many radial nodes are present?

Answer type: number | unit: radial nodes

Screenshot: quantum-stage-4-molecule



Screenshot: quantum-stage-4-question



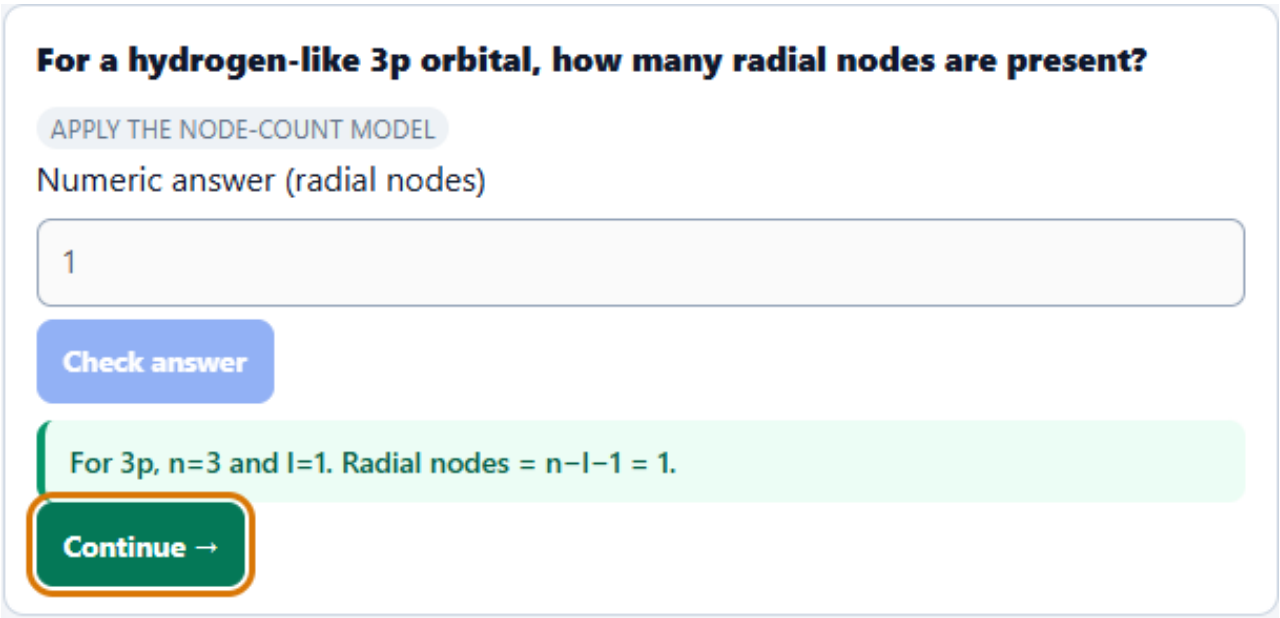
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Identify n and l from 3p.

Hint 2: Subtract l and one from n.

Explanation: For 3p, n=3 and l=1. Radial nodes = n−l−1 = 1.

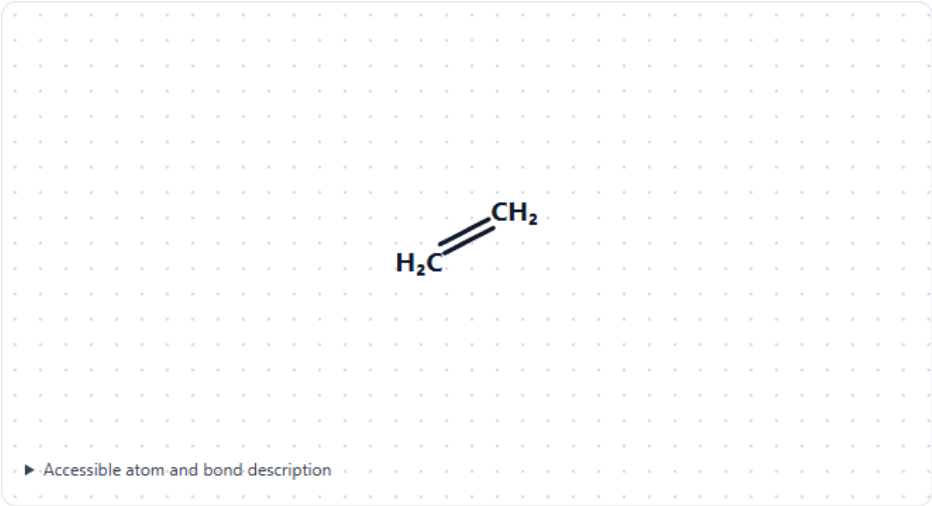
Screenshot: quantum-stage-4-feedback



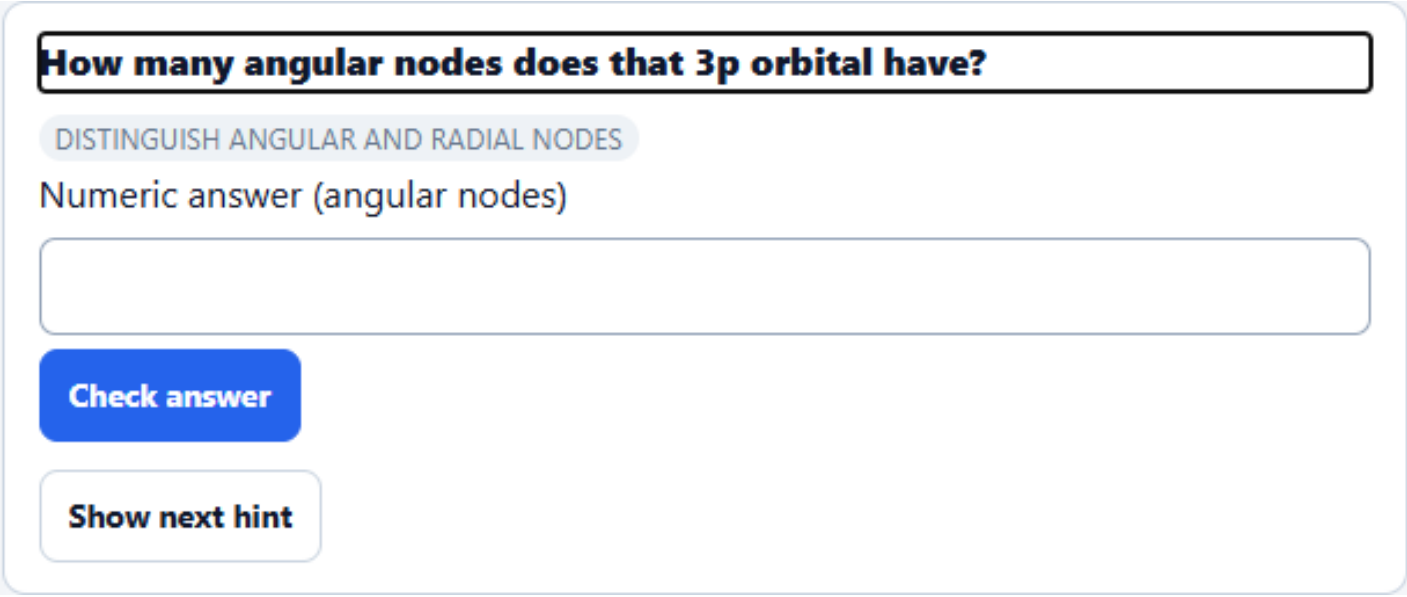
quantum:stage-5 | How many angular nodes does that 3p orbital have?

Answer type: number | unit: angular nodes

Screenshot: quantum-stage-5-molecule



Screenshot: quantum-stage-5-question



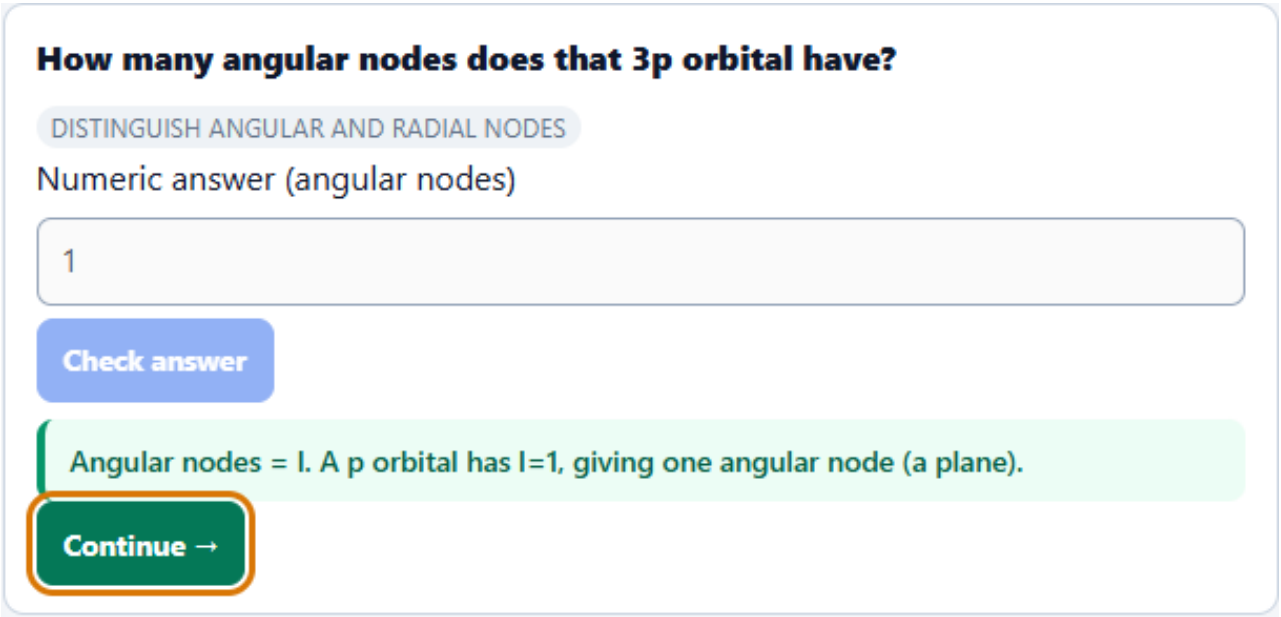
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the subshell letter rather than n.

Hint 2: The angular-node count equals l.

Explanation: Angular nodes = l. A p orbital has l=1, giving one angular node (a plane).

Screenshot: quantum-stage-5-feedback



quantum:stage-6 | In the usual multielectron-atom model, which listed n=3 subshell is lowest in energy?

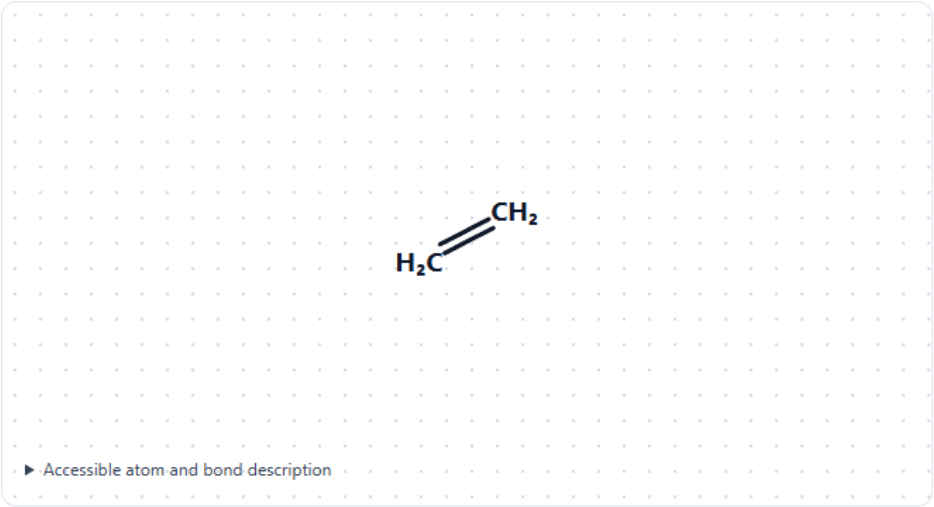
Answer type: choice

Option 0 (A): 3s

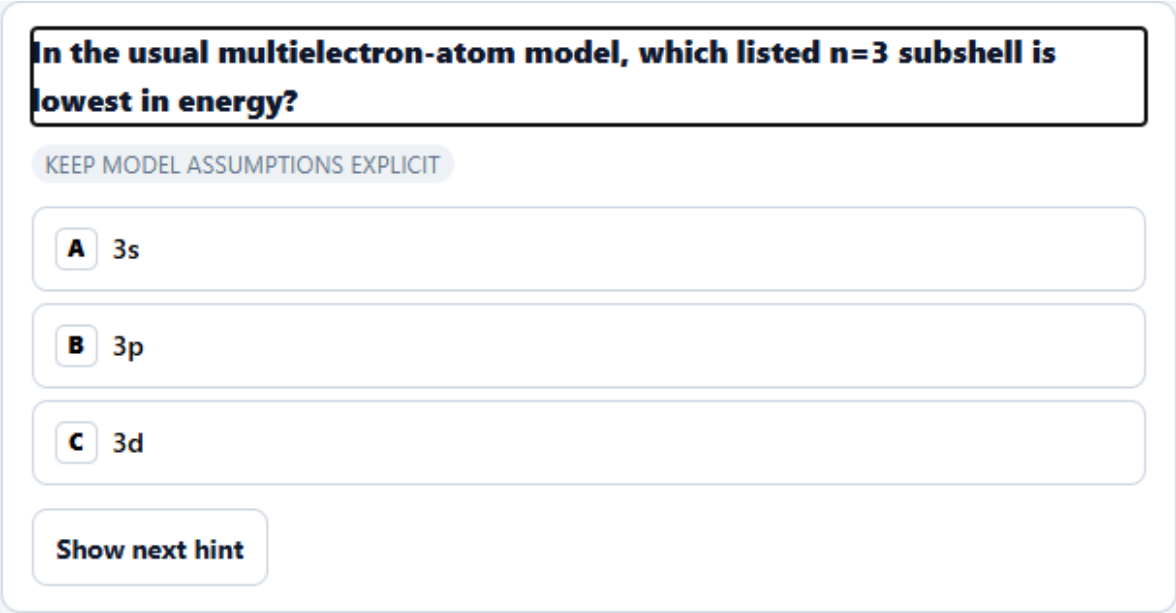
Option 1 (B): 3p

Option 2 (C): 3d

Screenshot: quantum-stage-6-molecule



Screenshot: quantum-stage-6-question



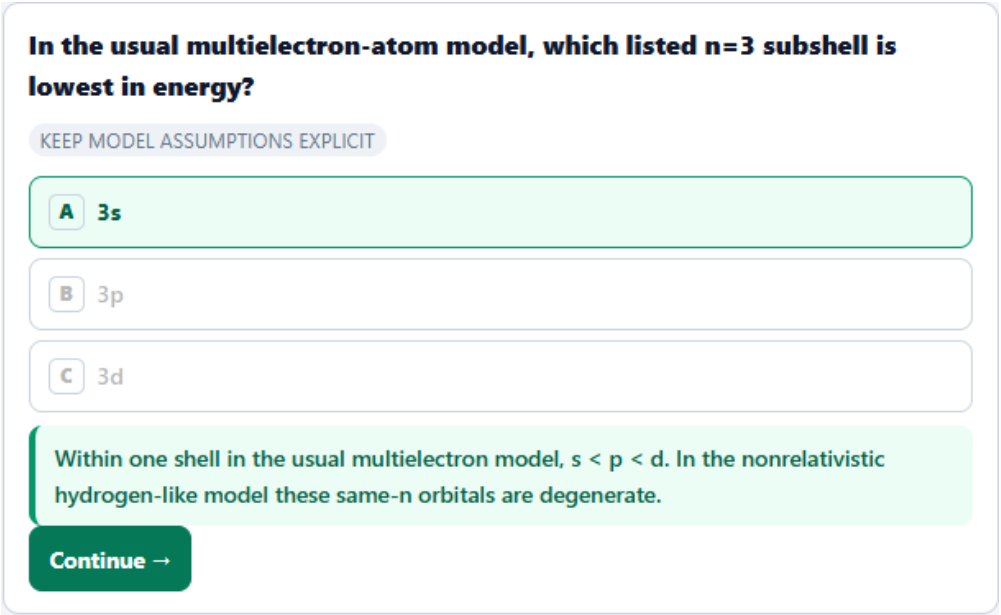
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Electron–electron interactions distinguish subshell energies.

Hint 2: Use the within-shell order s < p < d.

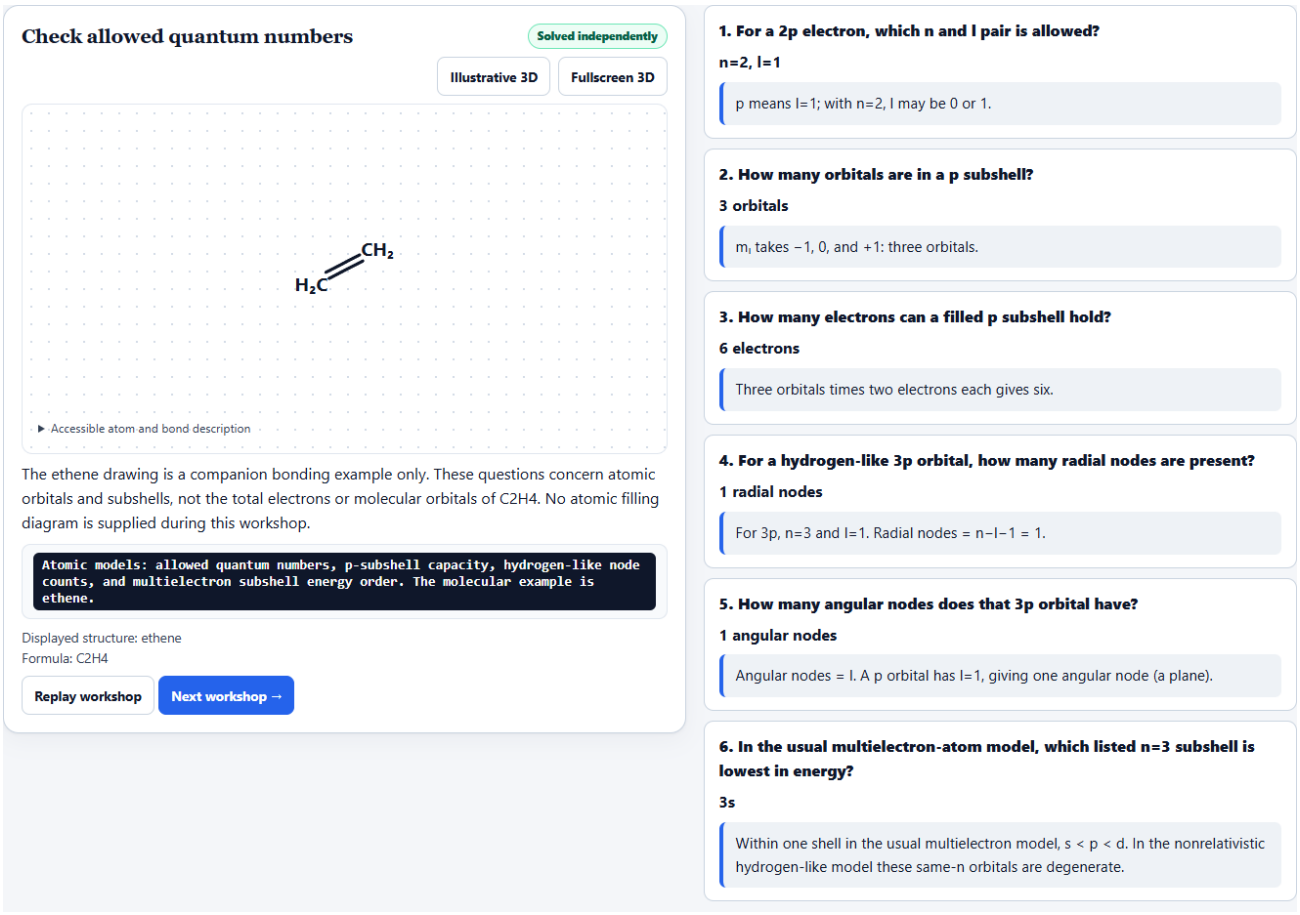
Explanation: Within one shell in the usual multielectron model, s < p < d. In the nonrelativistic hydrogen-like model these same-n orbitals are degenerate.

Screenshot: quantum-stage-6-feedback



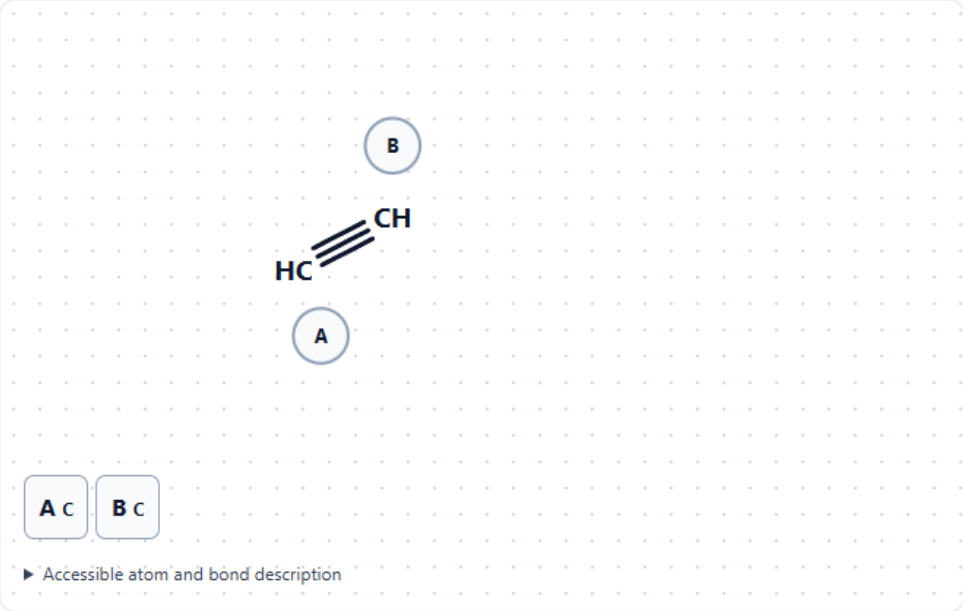
Completion summary: Atomic models: allowed quantum numbers, p-subshell capacity, hydrogen-like node counts, and multielectron subshell energy order. The molecular example is ethene.

Screenshot: quantum-complete



bond-count | Count all sigma and pi bonds

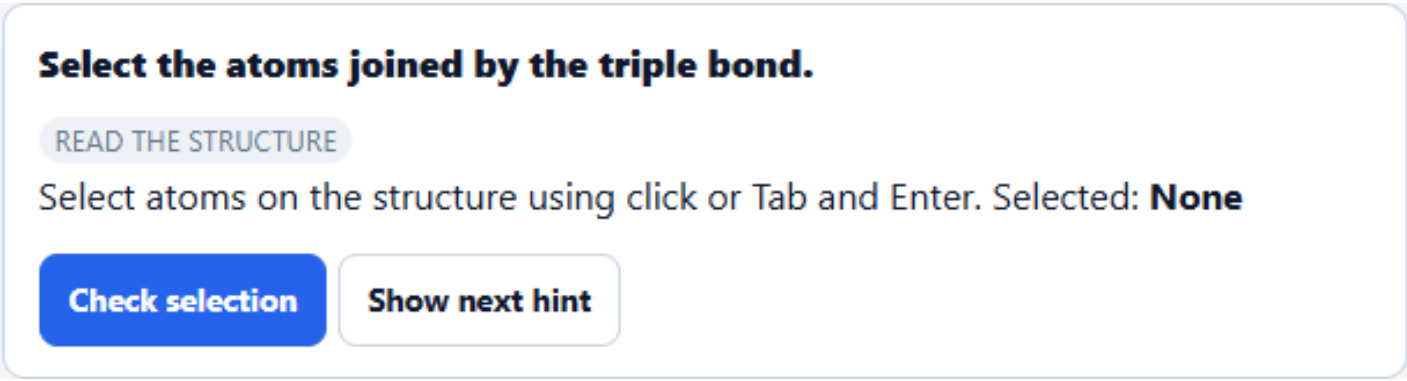
Screenshot: bond-count-stage-1-molecule



bond-count:stage-1 | Select the atoms joined by the triple bond.

Answer type: atoms

Screenshot: bond-count-stage-1-question



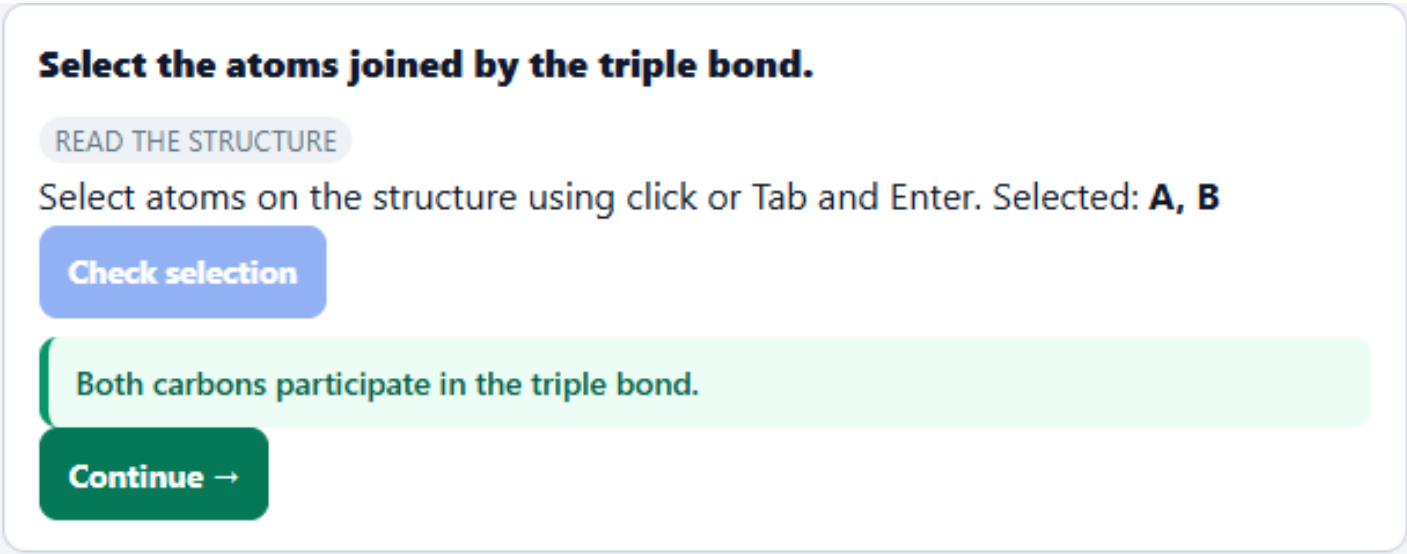
{"authorAnswer": [0, 1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 1]}}

Hint 1: Find the three parallel lines.

Hint 2: Select both ends.

Explanation: Both carbons participate in the triple bond.

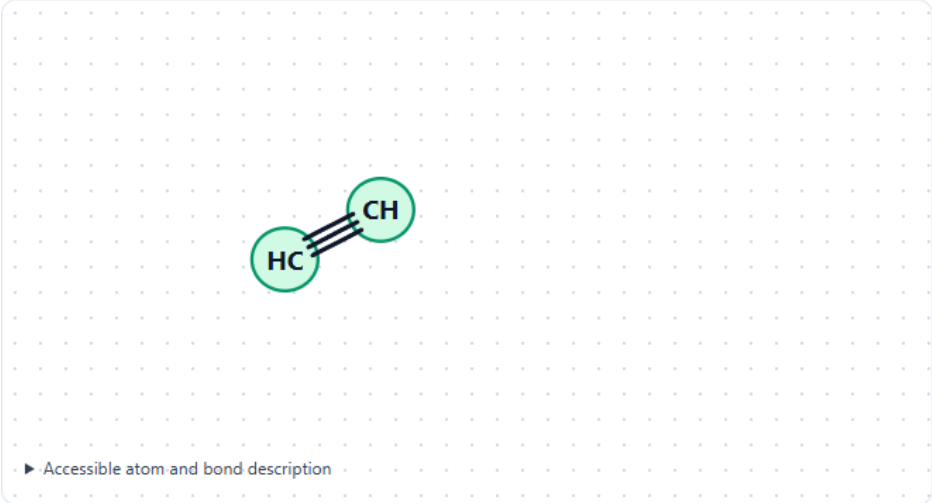
Screenshot: bond-count-stage-1-feedback



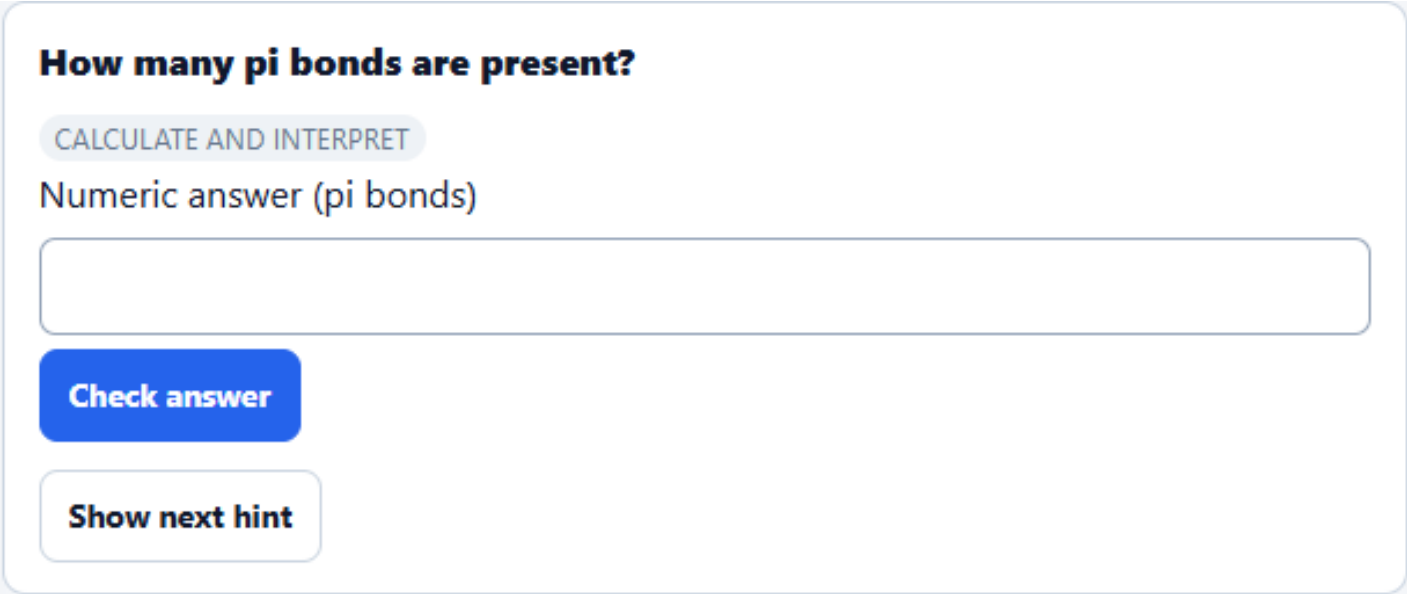
bond-count:stage-2 | How many pi bonds are present?

Answer type: number | unit: pi bonds

Screenshot: bond-count-stage-2-molecule



Screenshot: bond-count-stage-2-question



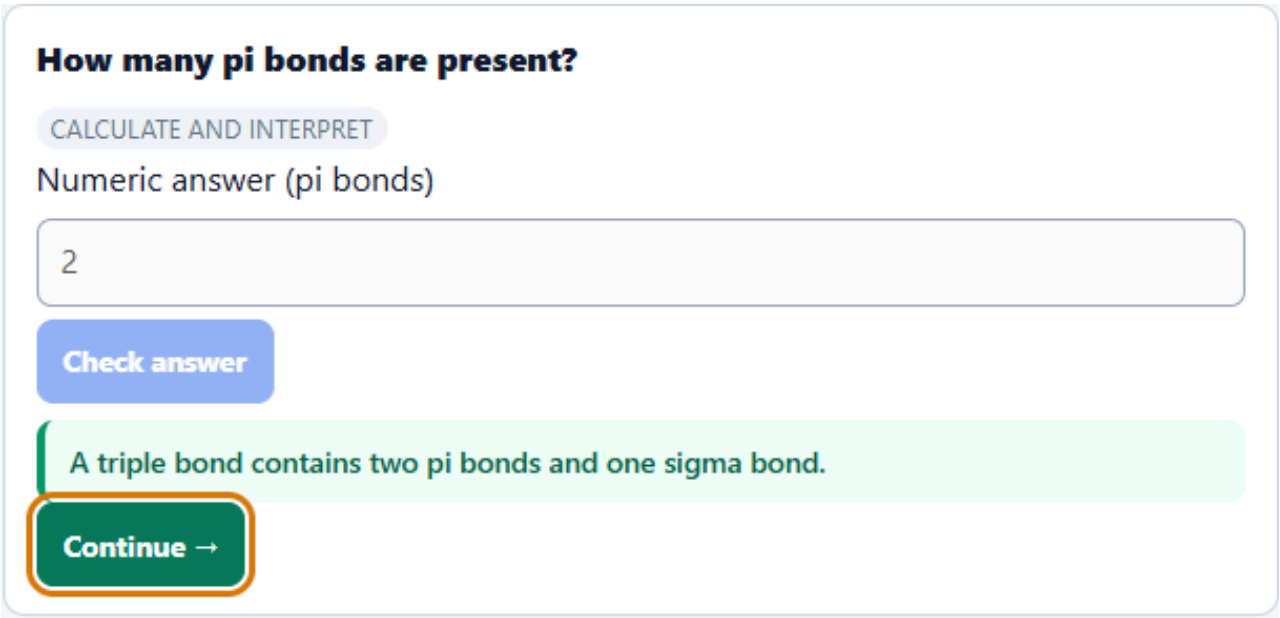
{"authorAnswer": 2, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Every bonded atom pair has one sigma bond.

Hint 2: The other two bonds are pi.

Explanation: A triple bond contains two pi bonds and one sigma bond.

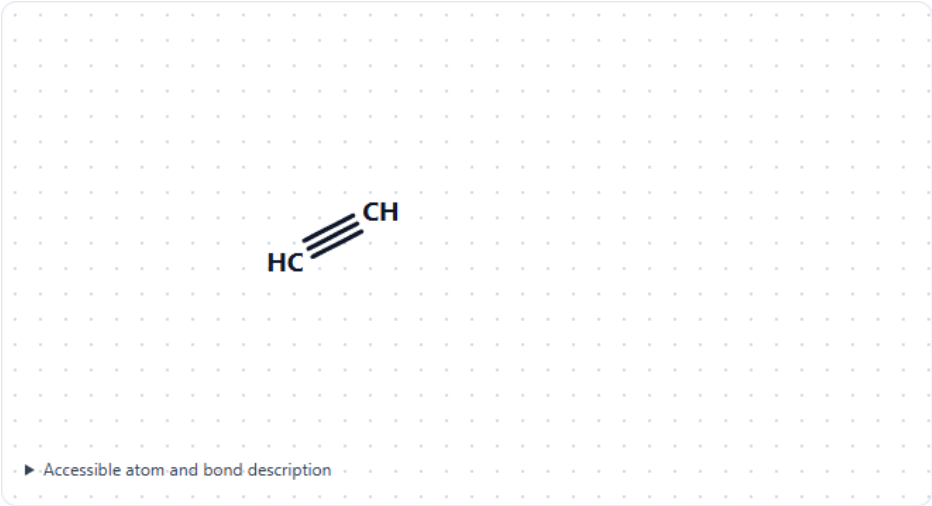
Screenshot: bond-count-stage-2-feedback



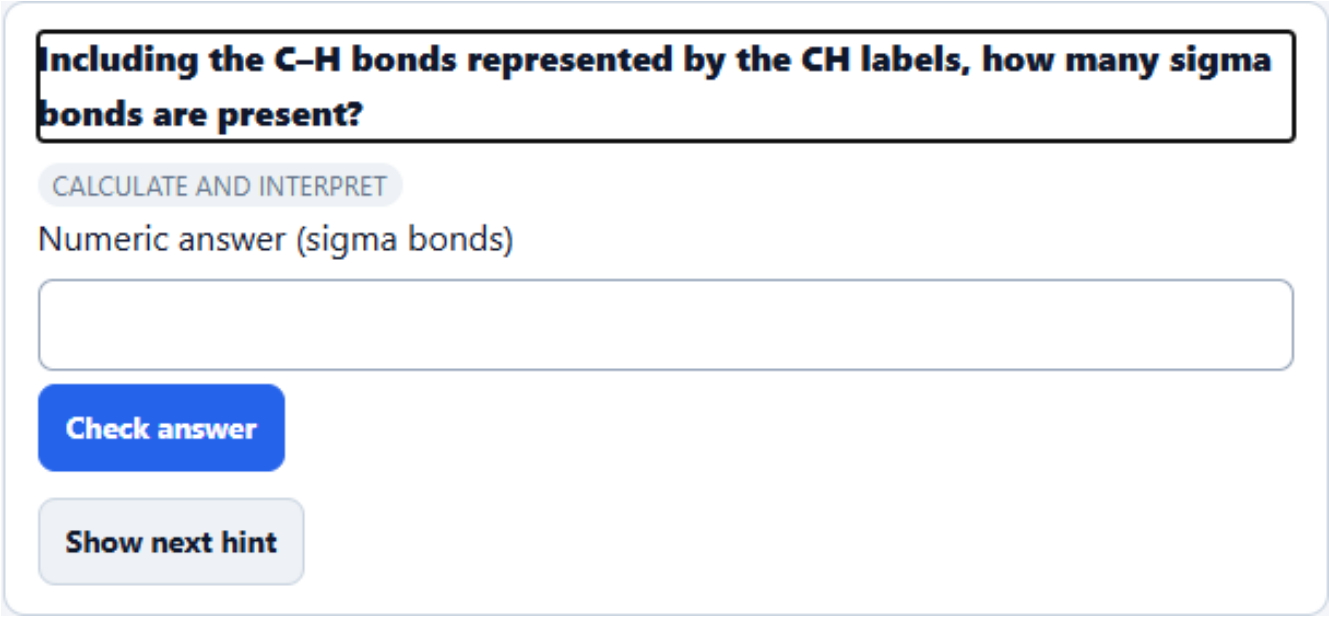
bond-count:stage-3 | Including the C–H bonds represented by the CH labels, how many sigma bonds are present?

Answer type: number | unit: sigma bonds

Screenshot: bond-count-stage-3-molecule



Screenshot: bond-count-stage-3-question



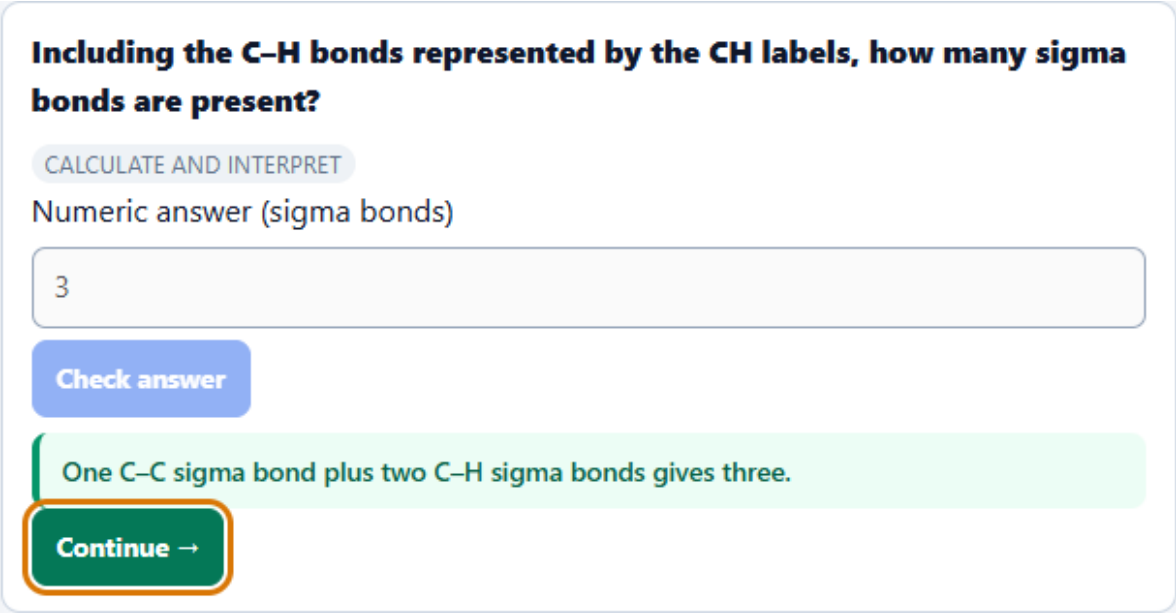
{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Each CH label includes one C–H single bond not drawn as a separate line.

Hint 2: Count the C–C sigma bond once.

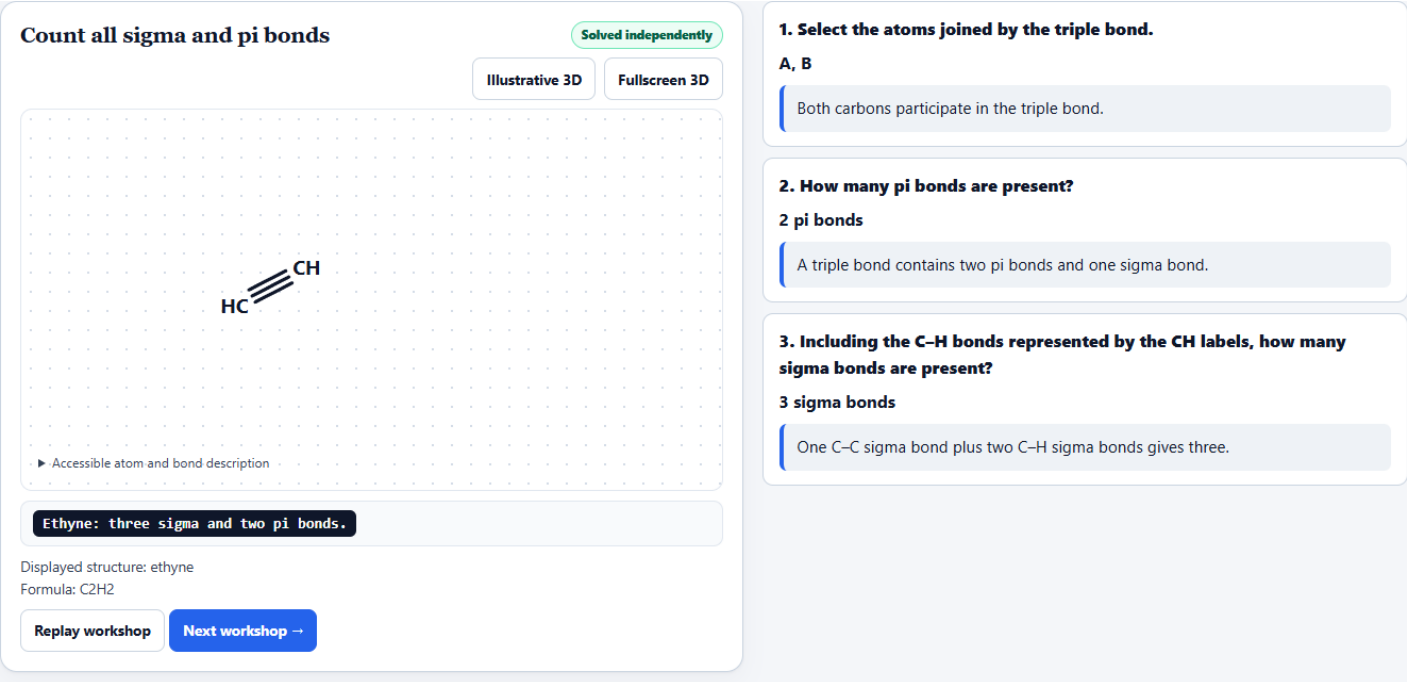
Explanation: One C–C sigma bond plus two C–H sigma bonds gives three.

Screenshot: bond-count-stage-3-feedback



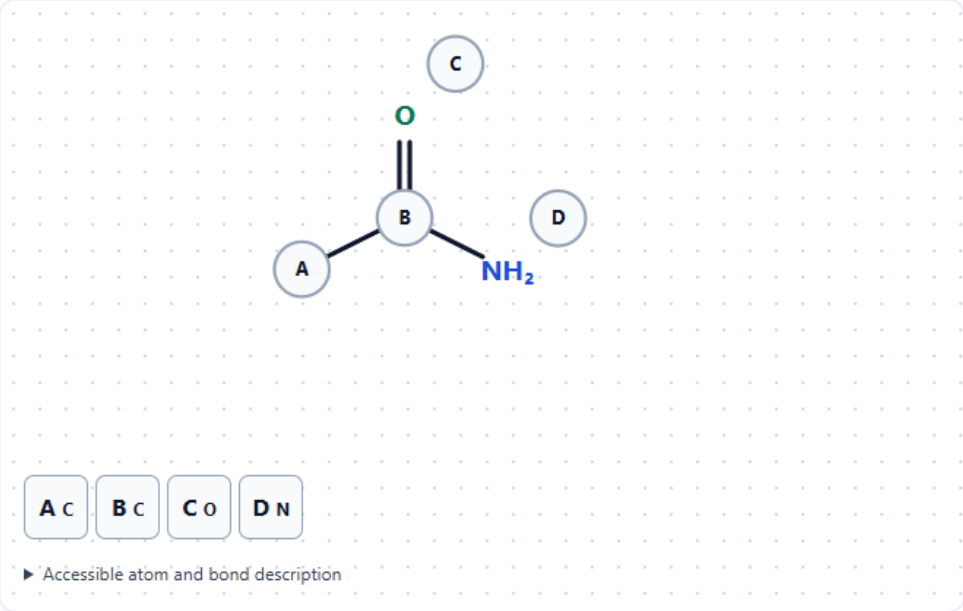
Completion summary: Ethyne: three sigma and two pi bonds.

Screenshot: bond-count-complete



amide-resonance | Use resonance to predict geometry

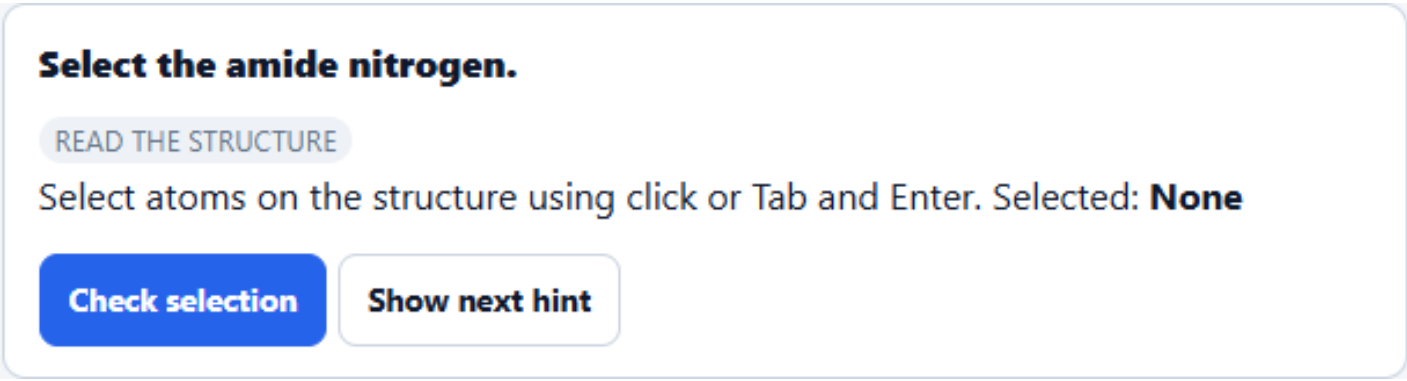
Screenshot: amide-resonance-stage-1-molecule



amide-resonance:stage-1 | Select the amide nitrogen.

Answer type: atoms

Screenshot: amide-resonance-stage-1-question



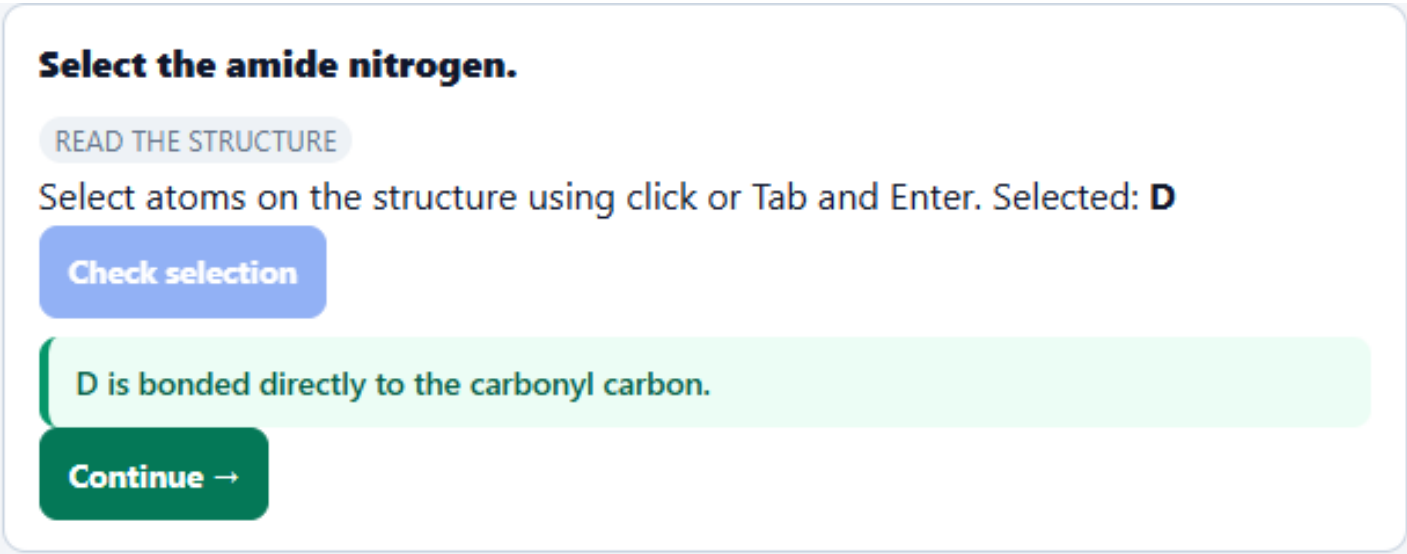
{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [3]}}

Hint 1: Read the N element label.

Hint 2: The amide N neighbors the C=O carbon.

Explanation: D is bonded directly to the carbonyl carbon.

Screenshot: amide-resonance-stage-1-feedback



amide-resonance:stage-2 | Why is the amide group approximately planar?

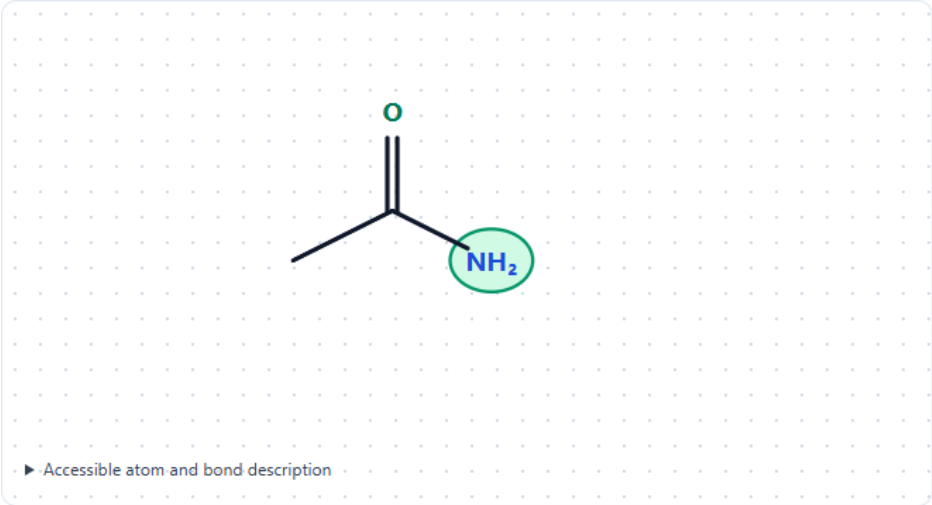
Answer type: choice

Option 0 (A): Nitrogen lone-pair conjugation with C=O

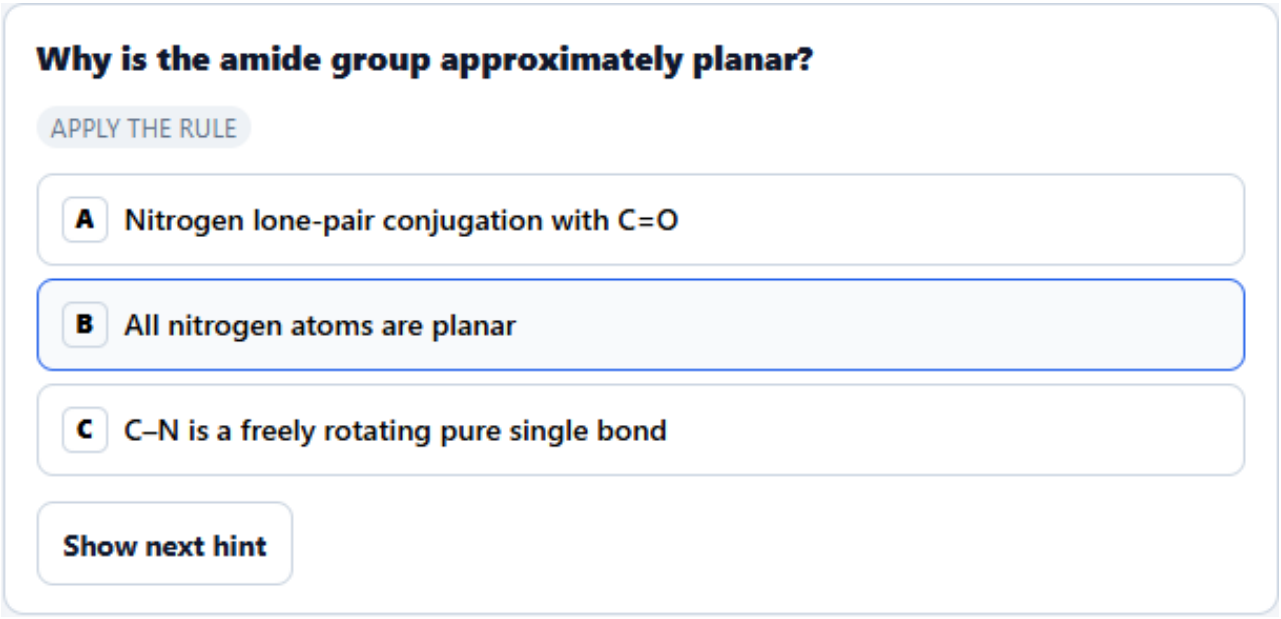
Option 1 (B): All nitrogen atoms are planar

Option 2 (C): C–N is a freely rotating pure single bond

Screenshot: amide-resonance-stage-2-molecule



Screenshot: amide-resonance-stage-2-question



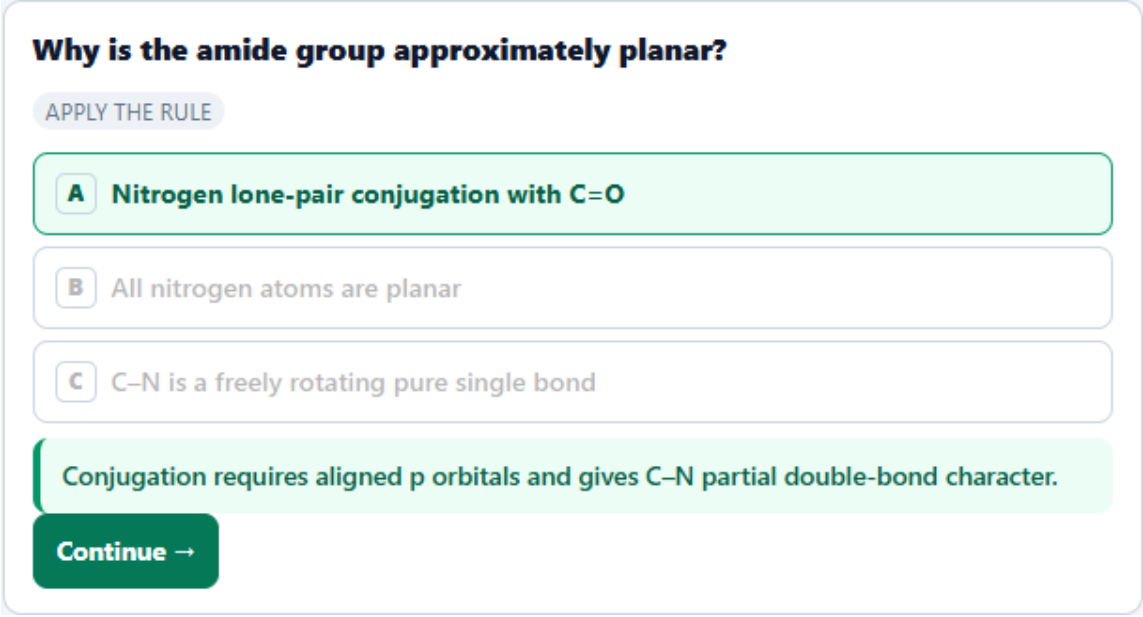
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Look for adjacent lone-pair and pi systems.

Hint 2: Delocalization requires orbital alignment.

Explanation: Conjugation requires aligned p orbitals and gives C–N partial double-bond character.

Screenshot: amide-resonance-stage-2-feedback



amide-resonance:stage-3 | For acetamide, resonance contributors differ in…

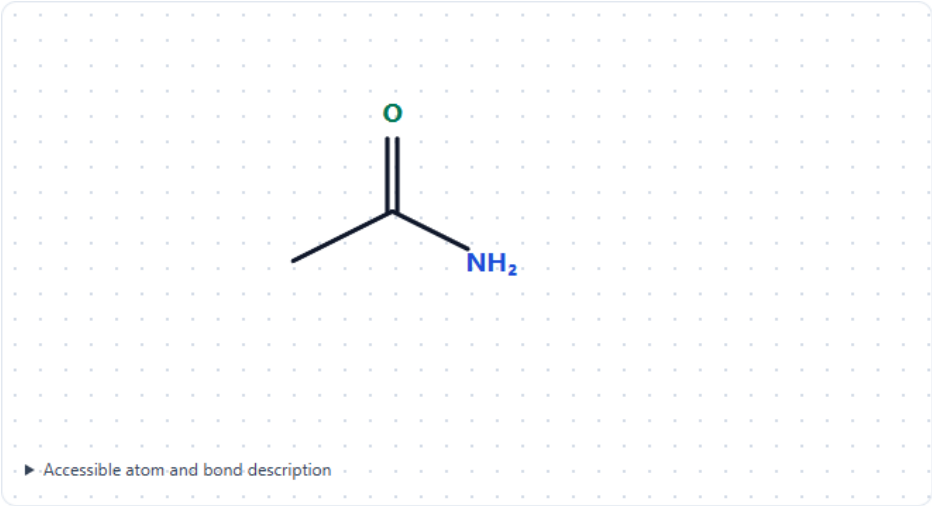
Answer type: choice

Option 0 (A): Positions of atoms

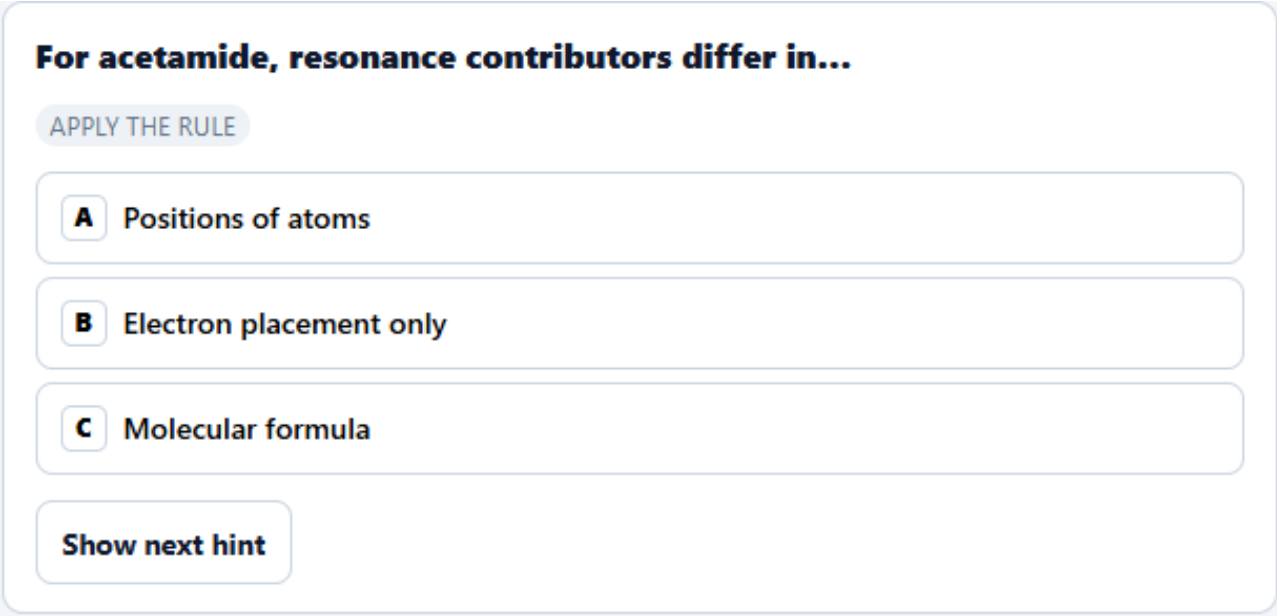
Option 1 (B): Electron placement only

Option 2 (C): Molecular formula

Screenshot: amide-resonance-stage-3-molecule



Screenshot: amide-resonance-stage-3-question



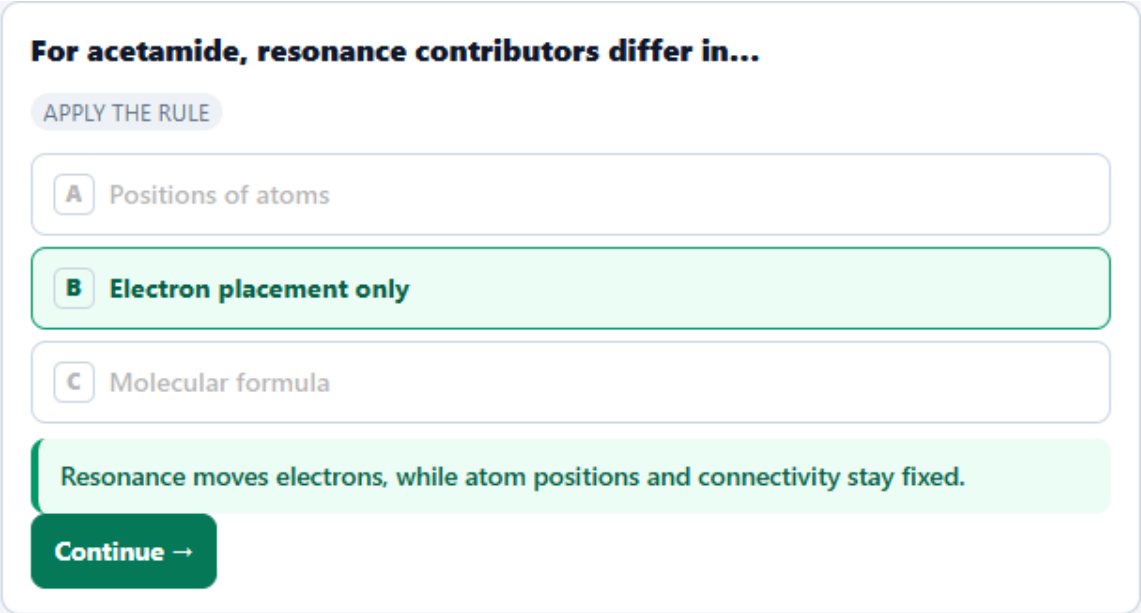
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish resonance from isomerization.

Hint 2: The sigma framework is unchanged.

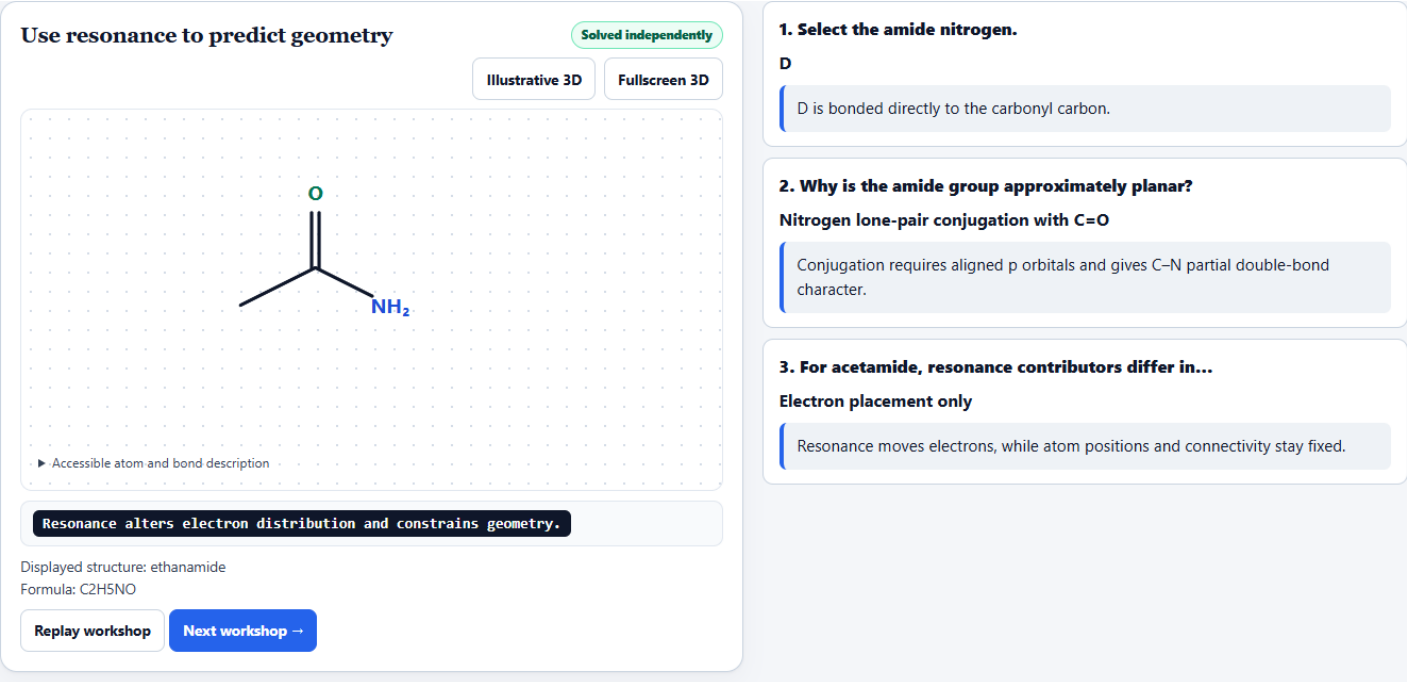
Explanation: Resonance moves electrons, while atom positions and connectivity stay fixed.

Screenshot: amide-resonance-stage-3-feedback



Completion summary: Resonance alters electron distribution and constrains geometry.

Screenshot: amide-resonance-complete



Molecular repository: author claims to verify

ethene

Author claims: ethene | formula C2H4 | canonical composition C2H4 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2]]

ethyne

Author claims: ethyne | formula C2H2 | canonical composition C2H2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 3]]

acetamide

Author claims: ethanamide | formula C2H5NO | canonical composition C2H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

acetaldehyde

Author claims: ethanal | formula C2H4O | canonical composition C2H4O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=189.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

acetate

Author claims: acetate (one resonance contributor) | formula C2H3O2 | canonical composition C2H3O2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

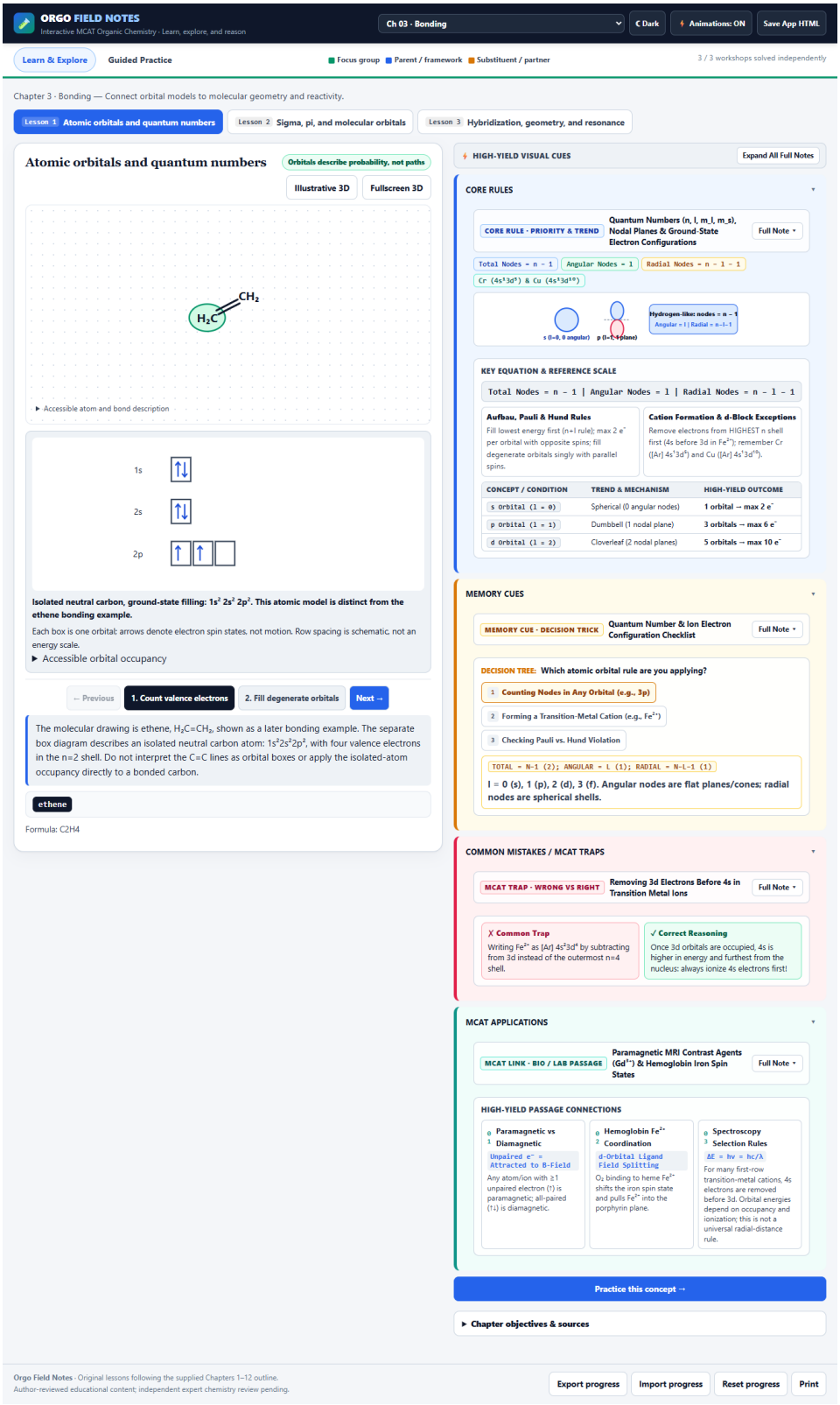
3 (D): O; x=175.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



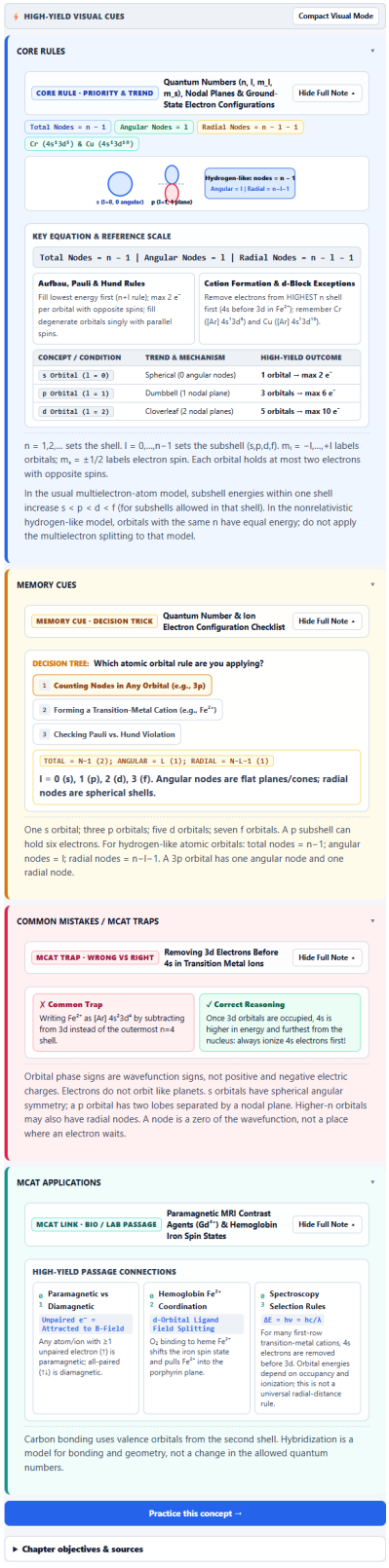
Screenshot: layout-mobile-light



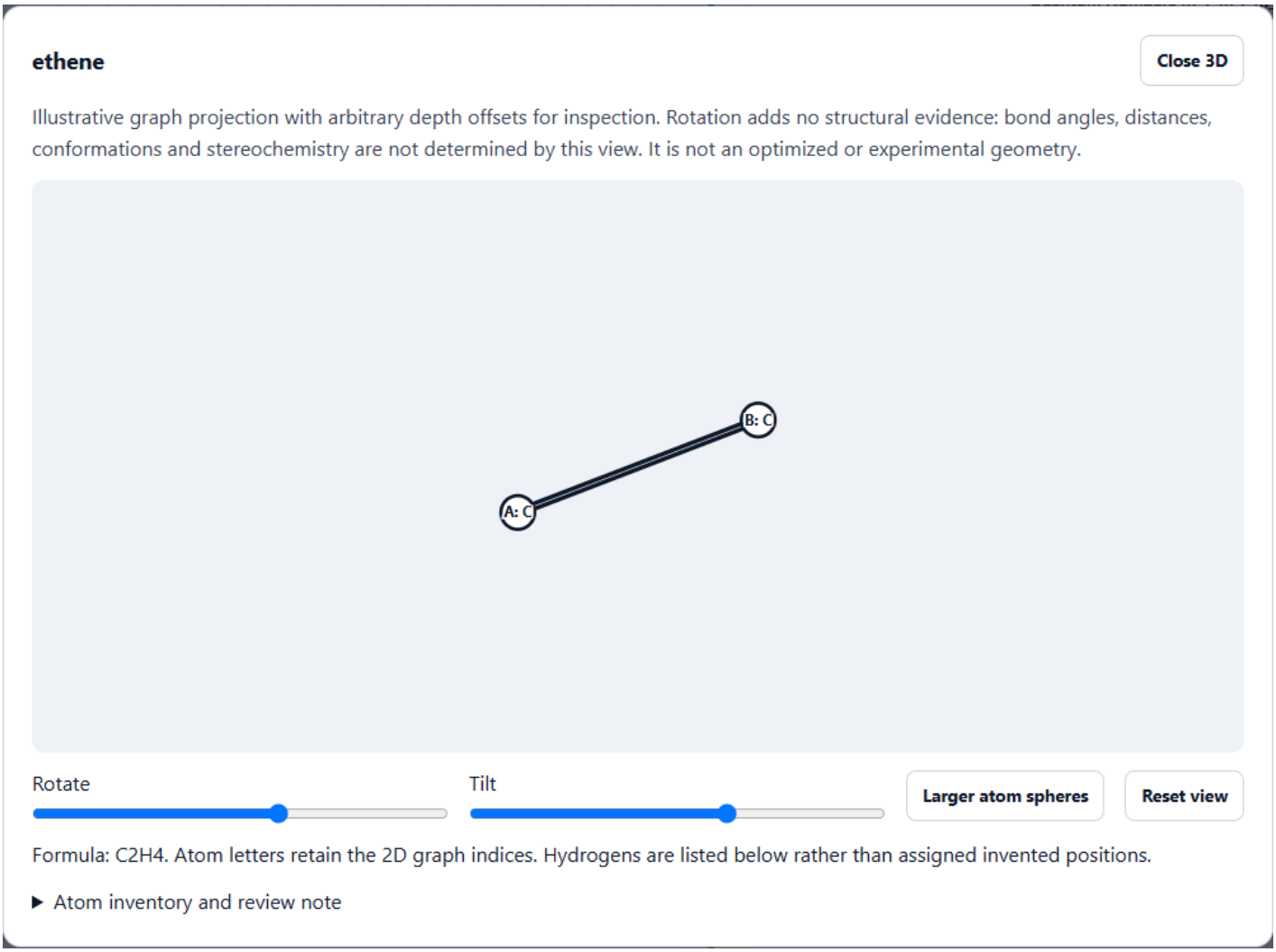
Screenshot: layout-mobile-dark



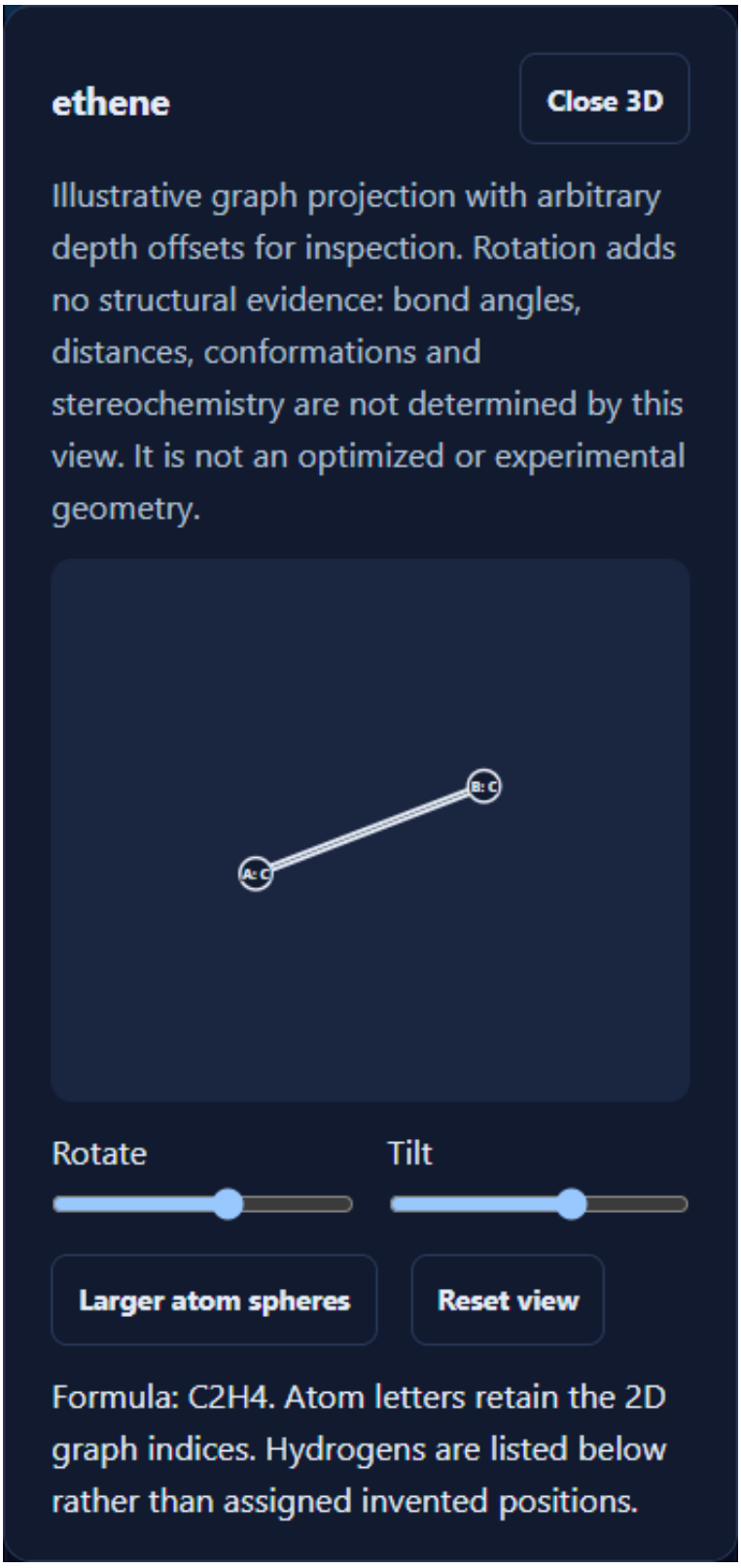
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

58 screenshots and 13 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.